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ARTICLE

Rifampicin tolerance classification is bounded by assay detection floor among clinical *Mycobacterium tuberculosis* isolates

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Abstract

Time-kill assays are the standard method for evaluating antibiotic tolerance, but constraints from starting bacterial density and the assay's detection floor are frequently overlooked, causing systematic misclassification of bacterial survival. We show how an unreported detection floor structurally distorts rifampicin tolerance classification in *Mycobacterium tuberculosis*.

We analyzed rifampicin tolerance against the assay's detection floor in 217 clinical *M. tuberculosis* isolates, isoniazid-susceptible and isoniazid-resistant alike, and derived two boundaries from the assay's own definitions: a reachability boundary, below which starting density cannot mathematically support the assigned log

reduction, and an identifiability boundary, below which a floor-level reading cannot be distinguished from a lower tolerance class. Both boundaries were then tested prospectively in *Escherichia coli*.

The standard classification proved structurally compromised. Thirty-three of 217 isolates lacked the starting density needed to physically reach their assigned 99.99 percent reduction, and none of them did. Eighteen isolates ended the assay exactly at the detection floor; a third were compatible with more than one tolerance class, admitting no unique classification. These floor-level outcomes tracked growth rate, and isoniazid-resistant isolates, seeded at a tenfold lower starting density than susceptible ones, were disproportionately exposed to this artifact. The *E. coli* experiment confirmed both boundaries outside *M. tuberculosis*: adjusting the plated volume alone assigned two conflicting tolerance labels to a single culture at six of 54 sample times.

A rifampicin tolerance classification made without reporting starting density and detection floor is mathematically uninterpretable, and isoniazid-resistant tuberculosis, read at the edge of that floor, is where the distortion concentrates.

Keywords: time-kill assay; limit of quantification; minimum duration for killing; antibiotic tolerance; colony counting; censored data; *Mycobacterium tuberculosis*

Introduction

Tuberculosis treatment runs for four to six months (1). Shortening it means finding drugs that kill *Mycobacterium tuberculosis* faster, and that choice is made in a flask: the compounds that could shorten a regimen are picked out of in vitro killing experiments. The experiment is simple: grow the organism, add the drug, and at set times dilute a sample, spread it on a plate, and count the colonies, each standing for one bacterium alive when it was plated, and the counts over time are the killing curve.

Every clinical microbiologist already knows the trap in that last step. A culture taken after antibiotics have started is not evidence of sterility. It is why blood cultures are drawn before the first dose. A drug suppresses growth, injures cells sublethally, carries over into the medium, and drives the population beneath what the plate can see; the plate then reports absence, and absence is not what happened. The rule is old, it is correct, and nobody disputes it.

The plate also has a hard floor, and the pipette sets it. The smallest positive result is one colony, and what one colony means depends on how much you spread. Plate 100 μ L and one colony is 10 bacteria per mL. That is the floor. Anything below it comes back as no growth, which is not the same as nothing there. Write *L* for that floor and *No* for the starting density.

Tuberculosis makes the old rule expensive, because the phenotype invoked to explain why treatment takes so long is itself defined by a duration. That phenotype is tolerance: the capacity of a genetically susceptible population to survive an exposure that should kill it. These bacteria are not resistant. The drug still works on

them. They just take a long time to die, and that time is the measurement. A phenotype defined by a duration is defined by exactly the observation the old rule warns about.

The field's response was to engineer around the warning rather than ignore it. Rather than asking whether a culture went negative, the modern definition asks how many logs the population fell: the minimum duration for killing, MDK, is the time to a specified fractional reduction (90, 99 or 99.99 per cent) and it is proposed as the tolerance counterpart to the minimum inhibitory concentration (2, 3). The choice of a *fraction* is the whole point. A ratio to the starting population is scale-free: if the count falls in a straight line on a log scale at rate b , the time to a q -log reduction is q/b , and the starting density cancels. By construction, MDK cannot be contaminated by how much culture went into the tube. That is the property it was built to have.

Two things are worth separating before going further. As defined by Brauner and colleagues and codified in the field's consensus guidelines, MDK is not a plate-count metric: it was first measured from presence or absence of survivors in microwell arrays of about a hundred cells, a design chosen specifically to avoid dilution plating, and the guidelines say how the smallest detectable count should be established (3, 4). What this paper examines is the form the tuberculosis field adopted: log-reduction endpoints computed from plate counts and most-probable-number series against a measured starting density. The floor problem belongs to that implementation, not the definition, and every deposit here uses it.

The property holds in the definition. It does not hold in the measurement, and the gap between the two is what this paper is about. To show a four-log kill you have to be able to see four logs down. A q -log reduction can be estimated only if q logs are visible, and what is visible is bounded below by the floor. So an isolate has

$$\text{headroom } h = \log_{10}(N_0 / L)$$

and no experiment can demonstrate a reduction deeper than h , however completely the drug worked. Where $h < q$ the endpoint is unreachable before the drug is added. That is the first boundary, and we call it reachability.

It gets worse when the last count lands on the floor. The fraction written down is then L/N_0 : the floor divided by the starting density. It is not a measurement of how many survived. The drug has dropped out of it, and what is left is the pipette and the inoculum. The metric designed to be inoculum-independent becomes a pure inoculum readout precisely at the depth where tolerance is scored. That is the second boundary, and we call it identifiability.

No prior treatment converts a floor-level reading into the classification rule it becomes. When the last count sits at L , the recorded fraction is L/N_0 , and the low, medium or high label is then a cut on starting density. That is the gap this paper fills. Two further failures have to be kept distinct from it. A duration ceiling (the assay stopped looking) is not a count floor, which is the plate being unable to see. And recording a below-limit flag is not the same as scoring a duration from the points so recorded.

This study asks how deep a log-reduction tolerance endpoint can be measured at all, and whether real isolates have been assigned tolerance phenotypes from outside that range. Two boundaries on the starting density

follow from the definitions, one deciding whether an endpoint is reachable and the other whether a floor-level reading identifies a class, and both are computed from quantities a time-kill protocol already records. Neither asks a laboratory to measure anything new.

Answering that question at all requires deposits that record the three quantities the arithmetic needs (a starting density measured before treatment, a time series, and either a plated volume or a stated floor), and most do not. Of 78 candidate time-kill datasets assembled for this study, 45 were inspected in full; of the 40 distinct literature deposits among them, **32 state no assay floor by any route**. That number is not a preamble to the analysis; it is the first result, and it is why the five deposits carried forward are five rather than fifty (Table S1).

If the hypothesis holds, a tolerance call reported without its starting density and its assay floor cannot be interpreted, and the phenotypes assigned in its name are, in part, a record of how the assay was set up.

Box 1. Four numbers that score an MDK. N_0 is the starting density. L is the smallest positive count the method can report: one colony in a plated volume v μ L is $L = 1\,000/v$ per mL, and an MPN series uses the lowest table rung (5, 6). Headroom is $h = \log_{10}(N_0/L)$. A q -log endpoint is legal only if $h \geq q$, which is the same as seeding at or above $L \cdot 10^q$. If the last reading sits at L , the recorded fraction is L/N_0 : an upper bound, not a measurement.

Culture	N_0 (per mL)	L (per mL)	Headroom	Legal endpoints	If the last reading is at L
Thin clinical MPN	23 000	23	3.00	90%, 99%, 99.9%; not 99.99%	Fraction 10^{-3} . Low and medium both compatible: the rule cannot choose.
Adequate clinical MPN	230 000	23	4.00	all four, including 99.99%	Fraction 10^{-4} . Only low is compatible.
100 μ L plate	10^5	10	4.00	through 99.99%; not a 5-log call	Fraction 10^{-4} . A 5-log MDK needs $N_0 \geq 10^6$.
10 μ L drop, same N_0	10^5	100	3.00	through 99.9%; not 99.99%	Fraction 10^{-3} . The pipette, not the isolate, removed one log.

The first two rows are the two starting densities that recur in the clinical deposit analysed below, and Section 2 reports what became of the isolates that began at each. The last two are why a written protocol does not standardise measurable depth: the plated volume sets L . To make both a four-log endpoint and a low class identifiable against $L = 23$ per mL, seed above $L \cdot \max(10^4, 1/c_1) = 230\,000$ per mL, or choose a shallower endpoint.

Results

1. The plate's floor sets how far down killing can be seen

Most-probable-number (MPN) readings take only the discrete values of an MPN table (5, 6), a fixed ladder of rungs, not a continuous scale. The day-5 column does not taper towards zero. It stops. Eighteen isolates sit at exactly 23 per mL in the 15-day panel and six in the 60-day panel, and nothing in the file lies below that (Fig. 1A). That is a floor, not a tail, and it is treated as one throughout.

Each isolate therefore carries a fixed budget of killing the assay can see. At 15 days of prior culture the starting densities run from 2,300 to 61,000,000 per mL (3.36 to 7.79 log₁₀), so headroom, the distance from the starting density down to the floor, runs from 2.00 to 6.42 log₁₀ (Fig. 1B, Table 1): both spans are the same 4.42 log₁₀.

Against that budget the three deposited endpoints behave very differently. Every isolate has room for a 90 or a 99 per cent reduction: 0 of 217 fall short in either case. **Thirty-three of 217 isolates (15.2 per cent) lack the headroom for the 99.99 per cent endpoint**, a four-log reduction that was unobservable for them before rifampicin was added.

All 33 are recorded as not having reached the endpoint, sitting at the **assay ceiling**: a census rather than an estimate, since the floor censors the count while the ceiling censors time, an isolate not yet at its endpoint after six days being recorded at the last day rather than the day it would have reached. Among the 184 isolates that did have four logs of headroom, 162 are also at the ceiling, 88.0 per cent.

2. The same floor reading was given different tolerance labels

The deposited tolerance level is a cut-off on the recorded surviving fraction, with no overlap between classes. The thresholds themselves are not deposited; recovered here, at day 5 and 15 days of prior culture, low tolerance covers fractions below 10^{-3} , medium 10^{-3} to 10^{-2} , high above 10^{-2} . Those cuts reproduce every usable class in the file, 203 of 203 (Table S2). Usable means the 203 of 217 isolates whose label is one of the three ordered classes; the other 14 carry the "MDR" category, which no ordered analysis can place (Table S3). What follows is an argument about the fraction the assay recorded, not about an independent clinical judgement.

Eighteen isolates ended at the floor. They share one reported floor-level observation, but their true final counts are unknown below the floor, so the assay cannot tell their surviving burdens apart down there. That is not the same as saying they were killed to the same degree. Their starting densities span 265-fold, from 23,000 to 6.1×10^6 per mL, so the same terminal reading implies a different range of compatible reductions in each: at least 3.00 log reduction for the isolate that began at 23,000, at least 5.42 logs for the one that began at 6.1×10^6 , and in either case the true reduction may be anything from that bound down to complete kill. What the deposit records instead is a single number per isolate, L/N_0 , and those recorded fractions span the same 265-fold, from

3.8×10^{-6} to 1.0×10^{-3} (Fig. 1C), because that is what the ratio is. They are upper bounds on survival, not measurements of it.

The labels those eighteen received differ accordingly. Six isolates that began at 23,000 per mL recorded a fraction of 10^{-3} and were classified **medium** tolerance; twelve that began at 230,000 or above recorded 10^{-4} or less and were classified **low**. A single cut on the starting density alone reproduces all eighteen (Table 2): once a reading is censored at the floor, the starting density fixes which label the published rule is able to return.

3. The affected isolates are named before the drug is added

Sections 1 and 2 counted isolates. The two boundaries derived in the Methods do more: given only the floor, the class thresholds and each isolate's starting density, they say in advance which isolates are affected, before any day-5 reading is used.

For this assay $N_{\text{reach}} = 23 \times 10^4 = 230,000$ per mL and $N_{\text{id}} = 23/10^{-3} = 23,000$ per mL. Applied to the 15-day panel, the first predicts the 33 isolates unable to reach the 99.99 per cent endpoint; the second predicts that of the eighteen isolates whose reading rests on the floor, twelve have only the lowest class compatible and six have more than one. Those are the three counts Sections 1 and 2 report.

Neither boundary predicts which isolates reach the floor; that depends on what rifampicin does. Both predict what a reading at the floor can be made to mean once it happens, and there the agreement with the deposit is exact, isolate by isolate: the set the algebra says has a single compatible class and the set the deposit records in the lowest class are the same set.

What is sharp is the boundary case. Six isolates sit at exactly 23,000 per mL, which is N_{id} itself. For them L/No equals c_1 exactly, the interval $[0, L/No]$ touches the threshold from below and spans two classes, and the strict inequality fails. Those six are precisely the six the deposit records as medium rather than low. A rule written with the inequality the other way would have called all six low and been wrong six times out of six. That is a test the arithmetic could have failed and did not, and it is what licenses using the boundary in advance: given a floor, a set of thresholds and a starting density, the affected isolates can be named before any reading is taken.

The empirical findings are therefore consequences of the definitions rather than properties of this deposit: any dataset reporting a starting density, a floor and a threshold classification can be checked against them.

4. The label tracks how fast an isolate grows, and its link to resistance cannot be separated from the inoculum

There are two candidate predictors of the deposited label, each tested at two culture ages and two endpoint depths, eight tests in all. Eight tests on one question will throw up a false positive on their own, so they are corrected together as one family. Two survive (Table S4).

The label tracks the **growth rate** of the isolate. Slower growth goes with a higher tolerance class, and it survives adjustment for starting density (OR 1.096 per day of time to OD 0.4, 95 per cent CI 1.032 to 1.164, $p = 0.0030$, $n = 202$; unadjusted OR 1.126, 1.062 to 1.193). The label also tracks **isoniazid resistance**, until starting density enters the model. Unadjusted, a resistant isolate has 2.32 times the odds of a higher tolerance class (1.30 to 4.12, $p = 0.0042$); adjusted, the odds ratio falls to 1.31 (0.67 to 2.57, $p = 0.42$), and the interval no longer excludes one. Starting density is the strongest term in the file: every ten-fold rise in the starting MPN halves the odds of a higher tolerance class (OR 0.480, 0.340 to 0.677, $p = 2.9 \times 10^{-5}$).

Why the adjustment does that is measurable. Isoniazid-resistant isolates go into this assay at 5.36 log₁₀ against 6.36 for susceptible isolates, ten-fold thinner (Mann-Whitney $p = 9.1 \times 10^{-14}$ (7)), and 22 of the 84 resistant isolates carrying an ordered label lack the headroom for the deepest endpoint, 26.2 per cent, against 9 of 119 susceptible, 7.6 per cent. They are one log short of observable killing before the experiment begins.

In that stratum the two associations part company. Growth holds and still survives correction (OR 1.116, 1.040 to 1.198, $p = 0.0024$). Resistance returns OR 2.11 (1.07 to 4.16, $p = 0.031$) nominally, but ranks second in the family of eight and fails its Benjamini-Hochberg critical value of 0.0125. The resistance association therefore leans on isolates that are not independent of one another, and it is already the association that starting density explains away; we report it as the weaker of the two on both counts.

5. One written protocol does not give every laboratory the same depth to look down

If the effect is a property of assay geometry rather than of clinical sampling, it should appear where one protocol, one strain and one stock are handed out deliberately. Van Wijk and colleagues (2023) ran exactly that exercise, sending a single stock of H37Rv to six blinded laboratories under one written protocol (8, 9). It does.

At ten times the minimum inhibitory concentration of moxifloxacin all six laboratories return a positive kill rate, spanning 5.2-fold, from 0.090 to 0.464 log₁₀ CFU/mL per day (Fig. 2A, Table S5).

The two orderings do not correspond. Rank the laboratories by starting density and the three lowest are exactly the three that recorded crossings, the three highest exactly the three that did not, without exception, a split with probability 1/20 under chance alone, reported as an ordering and not a test (Table S6). Rank them by kill rate and there is no such correspondence (Fig. 2B). Institute F has the highest starting density at 6.53 log₁₀ among treated flasks of this arm, kills faster than four of the other five at 0.223 log₁₀ per day, and records no crossing. Institute E returns the slowest rate of all at 0.090 and also records none.

6. The same boundaries hold in an experiment the framework never saw

Dubey and colleagues (2026) report amoxicillin-clavulanate against *Escherichia coli* in a hollow-fibre system (10, 11). Their Methods state 100 µL plated with counts per mL, so $L = 10$ CFU/mL is derived rather than inferred, and the file corroborates the derivation: all 229 genuine counts are multiples of ten, the smallest

exactly ten, while 69 further entries read as one, which no 100 μL plate can produce and which are the deposit's placeholder for below the floor.

Across the 20 cultures with a measured day-zero density, headroom runs from 4.85 to 5.45 $\log_{10}$ (Table S7). Endpoints at 1, 2, 3 and 4 logs are reachable for every culture; a 5-log endpoint is unreachable for 5 of 20, a 6-log endpoint for all 20. On data it was not built from, the framework locates the depth at which a sterilisation claim stops being demonstrable.

Two features of that deposit are the failure modes this framework is meant to catch. Its Methods give a nominal inoculum of 10^5 CFU/mL while the file measures a median of 1.215×10^6 , so taking *No* from the Methods rather than the data would have placed a full order of magnitude into every boundary. And its treated arm reads at or below the floor already at day zero while the paired control reads about 10^6 ; the file cannot distinguish a sample drawn after exposure from a placeholder or a transcription error, so the reading is flagged rather than treated as a measured time zero.

7. What the plate misses is not only what lies below its floor

Everything above is about the floor. One published observation in this organism shows that the floor is not the only reason a colony count understates the population, and the two failures must not be confused.

Evangelopoulos and colleagues counted the same mouse lungs three ways, colony count, a molecular bacterial load from 16S rRNA, and a most probable number, on their way to a different question (12). Under the deepest regimens the colony count reads zero in every animal: in three arms, 18 lungs in all, the plate declared every lung sterile while the most probable number on that same tissue ran from 1,450 to 18,700 per lung. The molecular load can be dismissed as nucleic acid outliving its owner; the most probable number cannot, because it is a culture, and growth in a dilution series requires an organism that is alive and culturable.

That gap is not the detection floor, and the arithmetic says so. For censoring to explain a plate reading of zero against 1,450 organisms in the same tissue, the colony count's floor would have to sit above 1,450 per lung, far less of the homogenate than any lung protocol plates. This is differential culturability, a described phenotype of *M. tuberculosis* with a literature of its own: resuscitation-promoting-factor-dependent bacilli dominate pre-treatment sputum and rise during chemotherapy (13), both dependent and independent differentially culturable populations are recovered from patients whose solid-medium cultures return nothing (14), and the mechanism has been reproduced experimentally, rifamycin action on ribonucleic acid (RNA) polymerase converting a starved population to over 90 per cent differentially detectable in vitro (15), the same drug class every arm of the Evangelopoulos regimen carries.

Reachability and identifiability are arithmetic on the plated volume; they would hold in an assay with perfect culturability. Differential culturability is biology, acts on the numerator rather than the floor, and nothing here derives or measures it. Neither depends on the other, and they push the same way: a count that is both floored

and culture-selective understates the surviving population twice over, and the mouse model that decides which regimens enter clinical trials reads out in colony counts, in exactly this region.

That deposit establishes no floor of its own. Five further *M. tuberculosis* deposits in the same screen do, and they are not reassuring. Across those five, sputum from pulmonary tuberculosis, axenic culture, an auxotrophic H37Rv panel and two drug-development series, with floors from 13 to 200 CFU/mL, **184 of 310 series could not have demonstrated a four-log reduction however completely the drug worked, and 240 of 310 could not have demonstrated five** (Table S8); one could not have shown four logs in a single one of its 123 series. None of these five carries the analysis above, and none is among the five deposits reanalysed in this paper; they say the reachability problem is not a property of one clinical panel.

8. One culture, two plated volumes, two tolerance labels

Every result above reads data somebody else generated for another purpose. This one reads an experiment built to try to break the two boundaries N_{reach} and N_{id} , with its predictions written down before a plate was counted. *Escherichia coli* ATCC 25922, the reference strain the standard names for quality control, against ciprofloxacin at ten times the concentration that stops it growing. Three seeding densities, two plated volumes, three flasks each, 216 plate readings.

The middle arm was seeded on purpose at the 5×10^5 per mL the standard specifies. The densities actually reached were 6.4×10^4 , 4.4×10^5 and 3.8×10^6 per mL as arm means, but N_0 is a measurement made once per culture rather than a setting the protocol supplies, so headroom belongs to a flask and everything below is reported per flask. At the 100 μL plating the low arm's three flasks have 4.19, 4.11 and 4.00 logs of headroom, the last exactly at the four-log endpoint. **At the standard inoculum plated at 10 μL , the deepest reduction the three flasks could report ran from 3.71 to 3.93 logs**, while those same three cultures plated at 100 μL ran from 4.88 to 5.05 (Fig. S5B, Table S9). Every flask is paired with itself across the two platings.

That much is arithmetic. So is the next step. When both platings of one sample rest on their own floors, the two fractions written down are L_1/N_0 and L_2/N_0 , so they can fall either side of a class threshold and give one culture two labels. **The quantity that could have come out zero is how often a real experiment lands in the window where those two fractions straddle the cut**, and it depends on where the cut is (Fig. S5C, Table S10). Put the cut at one per cent and the window is nearly shut. Put it at one in a thousand, where the clinical classification above cuts its lowest class, and six of 54 sample-times receive two different labels from one culture, one in nine. That cut is imported from the tuberculosis classification; it serves here as a ruler for the window's width, not a claim about *E. coli* biology.

Discussion

Every clinical microbiologist knows that a negative culture taken under antibiotic exposure is not proof of sterility. This paper reports that the number brought in to replace that judgement carries the same defect, and once it is a number, it is harder to see.

The standard is not silent on that choice, and this is the paper's sharpest finding rather than an awkwardness for it. M26-A's own section 1.3.2.5, *Volume Transferred*, ties the volume to the endpoint with a counting rule: after the defined 99.9 per cent killing, at least ten colonies must remain (16) (for a standard inoculum of 5×10^5 per mL that means at least 20 μ L streaked, so a 10 μ L streak fails the rule), and section 3.1 separately requires the smallest detectable count to be established by serial dilution. A guideline written in 1999 already required both numbers this paper asks for. The deposits reanalysed here do not meet it, not for lack of care but because the endpoint moved: M26-A nowhere contemplates a four-log call, and the volume rule did not travel with the deeper endpoints.

Take the assay's contribution out and what is left makes biological sense. Growth state predicts the tolerance class before and after adjustment for starting density (isoniazid needs KatG and active cell-wall synthesis (17), rifampicin needs transcription (18)), and a population that is not dividing offers less of what these drugs act on. The classification is reading physiological state, not drug susceptibility.

That axis is the one tuberculosis drug development already leans on. Killing assays of this design rank regimens before they enter trials, and tolerance followed in patients is now being read as within-host evolution (19). If a tolerance classification is partly an inoculum measurement, conclusions drawn from it, that one regimen sterilises faster than another, or that a patient's isolates are becoming more tolerant over therapy, are in part conclusions about how the assay was set up. The correction applied here does not remove the biology; it aims it at the right target. The phenotype that survives the correction is growth state, real, targetable and directly measurable, unlike a fraction read off a floor.

The prospective experiment has a limitation of its own, and it is the one a reader of this paper should raise first. Its samples were diluted in saline and plated; they were not washed, filtered or treated to inactivate the drug, and no carryover control was run. Antibiotic carried onto the plate suppresses colony formation without killing anything, the same class of counting artefact this paper is about, and at the shallow dilutions a culture near its floor is read on, dilution is the only thing containing it. The design is conservative in this respect rather than clean: the larger plated volume carries ten times more drug to the plate, so carryover would shrink the very difference between the two platings that Section 8 reports, not manufacture it. But the experiment cannot exclude carryover, and a repetition should include the control that would.

Several further limitations bound what the analysis can establish; each points at the experiment that would settle the question, and the ones bearing on the claims made are given here, with the rest in the supplement.

This paper has already acted on the first rather than merely declaring it. Flasks from one laboratory share a starting culture; repeat isolates come from a patient already counted. Every conclusion from the two primary deposits was recomputed at the level where the observations really are independent, listed one line at a time rather than summarised (Table S6): twenty survive unchanged, six survive with much wider uncertainty, five do not survive and **have already been removed from the text above**, and one is withdrawn because the null it was tested against was already false before any data were seen. The ledger is printed so a reader can check the dropping was done rather than promised.

The 15-day and 60-day panels of the clinical deposit contain the same isolates. The comparison between panels is therefore not independent, and it bounds the evidence for a difference rather than establishing it.

Choose the depth of the endpoint to fit the headroom actually available, rather than densifying the inoculum merely to buy more. The inoculum this needs can be worked out before the experiment: seeding above $L \cdot \max(10^q, 1/c_1)$ makes both failure modes impossible, the endpoint that was never reachable and the last count that lands on the floor; for this assay that is 230,000 per mL. Seeding higher also changes the experiment, because a denser culture is a different physiological state, and physiological state is what the tolerance label tracks, so where the deep endpoint is wanted, lower the assay floor instead; where the headroom will not carry it either way, the shallower endpoint is the honest one, and growth state belongs on the report whichever is chosen. This is a design decision that is currently not being made.

Treat an endpoint at the floor as a bound, not a value. Analysed as a measurement it manufactures differences between isolates whose final viable burdens the assay could not tell apart, and the same censored reading stands for a different demonstrated kill at every starting density.

Fix the inoculum, or adjust for it, before comparing tolerance across groups. The comparison that motivated this work, resistant against susceptible isolates, is confounded by a ten-fold difference in starting density, and nobody knows why that difference is there. Until it is understood, group comparisons of tolerance in clinical isolates should carry the starting density as a covariate.

Box 2. The minimum information a time-kill assay must report. Five fields that make a reported tolerance endpoint recomputable by a reader. Each is produced by the experiment as run; none asks for new apparatus. What they cost is visible from the other side: of the 78 candidate time-kill datasets assembled for this study, 45 were inspected in full, and of the 40 distinct literature deposits among them, 32 state no assay floor by any route, so most published endpoints cannot be recomputed against their own measurement.

#	Field	What to report	What fails without it
1	Starting density	N_0 with its method (viable count or MPN value, not an optical density)	the endpoint is a fraction of N_0 ; OD-derived densities disagreed with the measured ones by up to 1.53-fold within one culture here
2	Assay floor	L as a value, or the plated volume / dilution design that fixes it	nothing deeper than $\log_{10}(N_0/L)$ was ever visible; without L a failed endpoint is arithmetic
3	Censoring convention	what is written when nothing grows: zero, the floor value, ND, or the lowest table rung	the convention decides whether a floor reading is a bound or a measurement
4	Endpoint rule	the q in MDK _q and the interpolation between sample times	one kill curve returns different durations under different rules
5	Physiological state	culture age or growth phase at sampling, as a recorded value	the label tracks state; unrecorded, state is confounded with every group compared

A report carrying all five can be audited against its own floor, which is the test applied to every deposit in this article.

A time-kill guide already exists and this five-field minimum does not replace it: ASTM International (the American Society for Testing and Materials) guide E2315 sets out how to run and report a time-kill procedure in general (organism, media, neutralisation, raw counts) for antimicrobial products broadly, and none of its reporting requirements is L , headroom or a censoring convention specific to a duration endpoint cut on a fraction of the starting count (20). The five fields above are additional to that guide, not a substitute for it, and apply wherever a killing curve is read against an assay floor to score how long killing took rather than only whether it happened.

Conclusion

A plate that grows nothing under antibiotic has never been evidence that the flask was sterile, and the field knows it. The remedy adopted was to stop asking whether the culture went negative and to start asking how far it fell, expressed as a fraction of where it began so that the starting culture could not matter. We find that the fraction stops being a fraction once the population reaches the floor. What the deposit records from that point is L/N_0 , which moves with the starting density and not with the surviving population: in this deposit, for fifteen per cent of isolates at the depth the classification uses, and for the eighteen isolates whose recorded label a single threshold on their inoculum reproduces exactly, because for twelve of them no count below the floor would have changed which class the rule could return.

The consequence is that a tolerance call made without its headroom cannot be told apart from an inoculum measurement, and that the phenotype which does survive the correction is physiological state rather than drug susceptibility, not that tolerance is an artefact. Report the starting density and the assay floor. Match the depth

of the endpoint to the range actually available. Treat a reading at the floor as a bound. Do those three things and a tolerance classification would mean the same in one laboratory as in the next. Until then, some of what the field calls tolerance cannot be separated from a record of how much culture went into the tube.

What this study adds is a way to see the problem before the plates are poured: two boundaries computed from quantities a time-kill protocol already records, a classification that refuses rather than guesses when the assay cannot decide (Table S11), a five-field minimum that makes any killing endpoint recomputable (Box 2), and a small tool that returns the endpoint a planned assay can support. None of it asks a laboratory for a new measurement; the cost is five numbers written down. The return is a tolerance phenotype that means the same in one laboratory as in the next, and that can bear the weight regimen research already puts on it.

Materials and Methods

Study design and data sources

This study combined mathematical derivation, analysis of five public datasets, and a prospective *Escherichia coli* experiment to examine how assay floors constrain antibiotic-tolerance endpoints. We assembled 78 candidate time-kill datasets from the published literature (21 August – 8 September 2026) and screened 45 of them field by field for the three quantities the two boundaries below require: a pre-treatment starting density, a longitudinal series of viable counts, and either a plated volume or a stated assay floor. Every inclusion and exclusion is recorded in a manifest deposited with the analysis code.

Forty-one distinct records were inspected, one of them this paper's own prospective experiment. Of the 40 remaining literature deposits, 32 reported no assay floor by any route: no limit of detection, no limit of quantification, and no plated volume from which one could be derived. Eighteen deposits carried enough information to compute both boundaries at all, and five carried the complete set of fields the full analysis requires (Table 3, Figure S4). The full pipeline, from this screen through the prospective test of both boundaries, is summarised in Figure S6.

These 78 candidates are a documented sample of the literature, assembled without a registered search protocol; the two boundaries are derived from definitions rather than estimated from the corpus's size, so a single deposit with a stated floor is enough to demonstrate them.

Four datasets supported the empirical analyses. **Clinical isolates with a deposited tolerance**

classification: 217 *M. tuberculosis* isolates assayed under rifampicin, each carrying a minimum inhibitory concentration, minimum durations for 90, 99 and 99.99 per cent killing at 15 and 60 days of prior culture, most probable number readings at days 0, 2 and 5, a growth-rate proxy, isoniazid susceptibility with the resistance mutation where present, and the authors' own tolerance level for each isolate (Vijay et al. 2024; eLife 93243

supplementary file 2) (21). Of the 217 rows, 43 are follow-up isolates from patients already represented, so a baseline-only stratum is analysed separately.

The six-laboratory exercise: *M. tuberculosis* H37Rv from one stock, distributed with one written protocol to six blinded laboratories (van Wijk et al. 2023; figshare 19766083, CC BY 4.0) (8, 9). Moxifloxacin and isoniazid at one and ten times the minimum inhibitory concentration, three flasks per arm, each sample plated at four volumes: 100 µL quadruplicate, 10 µL as four drops, 10 µL as a single drop, and 2.5 µL as a single drop, giving three distinct floors spanning 1.60 log₁₀ within a single flask-visit (22). The file holds 2,775 readings; the analysis set, after dropping unusable values and pre-treatment visits, is 2,580.

Evolved clones: 126 *Escherichia coli* clones from a parallel evolution experiment under amikacin, each carrying an endpoint minimum inhibitory concentration and a persister fraction (Windels et al. 2024; Zenodo 10.5281/zenodo.7550302, CC BY 4.0) (23, 24). Six of its 595 surviving fractions are written as exact zeros with no named floor, so this deposit is one of two for which observability labels are refused.

Concentration-by-time grid: apramycin against *M. tuberculosis* at 128, 32, 8, 4 and 1 µg/mL, triplicate log₁₀ CFU at days 0, 3, 7 and 14 (Kaur et al. 2024; figshare 26462791, CC BY 4.0) (25, 26). The 1 µg/mL arm shows net growth rather than killing and is excluded from the dose comparison, leaving 4 to 128 µg/mL as the range compared (Section 6).

A fifth dataset was evaluated only after the boundaries, thresholds, and modelling choices had been fixed.

Validation deposit: amoxicillin- clavulanate against *Escherichia coli* in a hollow-fibre infection model, with 100 µL plated, measured day-zero densities for every culture, and below-limit readings retained as a placeholder (Dubey et al. 2026; article Source Data, CC BY 4.0) (10, 11). It was the only deposit found carrying all three fields the boundaries require, and no boundary, threshold or modelling choice in this paper was informed by it; it is used nowhere else in the paper except to confirm them (Text S3).

Each dataset's organism, drug, design and licence is summarised in Table 1; the complete screening manifest and exclusion reasons accompany the analysis code.

Prospective experimental validation

An *in vitro* experiment was designed to prospectively test both boundaries. Its design sheet, predictions and plate counts are one workbook, deposited with the analysis code, with every prediction written down before a plate was counted.

Escherichia coli ATCC 25922, the CLSI quality-control reference strain, was grown in cation-adjusted Mueller–Hinton broth and plated on Mueller–Hinton agar, incubated at 35 ± 2 °C in ambient air (16, 27). The ciprofloxacin MIC, determined by CLSI reference broth microdilution (27, 28), was 0.008 mg/L, within the CLSI quality-control range for this strain (28); the exposure was set at ten times it, 0.08 mg/L, chosen to kill deeply and fast enough that every arm reaches the floor within the sampling window.

Three seeding arms were set so the four-log endpoint is unreachable in one, marginal in the second and comfortable in the third: targets of 5×10^4 , 5×10^5 and 5×10^6 CFU/mL (the middle arm the CLSI-specified inoculum), reaching 6.4×10^4 , 4.4×10^5 and 3.8×10^6 CFU/mL as measured from day-zero counts: headroom of 4.11, 4.94 and 5.88 logs at the pooled 100 μ L floor, and 3.11, 3.94 and 4.88 logs at the pooled 10 μ L floor, so the lowest arm is reachable for a four-log endpoint only at the deeper plating. Three flasks per arm were sampled at 0, 1, 2, 4, 6 and 24 h; at each time, one dilution series was plated at both 100 μ L and 10 μ L in duplicate, giving 216 plate readings in all. Viable density was $N = C \cdot D \cdot 1000 / V$ for C colonies counted at dilution factor D and plated volume V μ L.

No carryover control was run. Because the 100 μ L plating carries ten times as much drug onto the plate as the 10 μ L plating of the same sample, any carryover effect works against the depth advantage the 100 μ L plating is reported to have, so the paired comparison in Section 8 is conservative with respect to it.

L is one colony in the volume plated, referred back to the undiluted sample. The two technical plates of a sample are pooled rather than averaged, so their volumes add and the floor is lower than a single plate's: 5 CFU/mL for a pooled 100 μ L pair against 10 CFU/mL for one plate, and the two platings differ by exactly one log by construction. Headroom, censoring at the floor and the three tolerance classes are applied unchanged from the definitions below; class labels use the thresholds of the clinical deposit above, $c_1 = 10^{-3}$ and $c_2 = 10^{-2}$, so the two experiments are scored on one rule.

Assay floors and censoring

No deposit analysed here reports a validated limit of quantification with a value. Throughout this paper L is therefore an **operational assay floor**: where a plated volume is recorded it is the **minimum reportable positive count**, one colony in that volume; where none is recorded it is inferred from the deposit and labelled as such. The term limit of quantification is used only in reporting what a source states.

In running text L is the **assay floor**, and that is the only short form used, so that "boundary" is left free for the two derived boundaries of the next subsection and never denotes a value. The event a time-to-event analysis records is correspondingly a first observed crossing below the assay floor. What L is in each deposit is tabulated deposit by deposit (Table S12).

Two censoring problems are handled separately. A **count below the assay floor** is left-censored: the population is somewhere below a known value. In the clinical deposit the floor is inferred at 23 MPN/mL from three signatures: it is the smallest value in the file, nothing lies below it in 1,932 readings, and the minimum shifts by exactly a factor of ten per culture age; the conclusions drawn from it hold across the full range the deposit's own MPN table admits (Table S13). A **crossing time not observed within the study** is right-censored: a flask that never falls below the floor contributes the information that its crossing time exceeds its last visit, and enters the survival likelihood as such rather than as missing.

A volume-derived plate floor bounds the true count from above, $[0, L]$; an MPN rung does not, since its likelihood extends well above the rung. Every identifiability verdict was therefore recomputed under the assay's own likelihood rather than the plate-sweep bound, confirming the same classification under both treatments (Table S14). Each below-limit flag in the six-laboratory deposit is judged against its own plated volume's floor rather than pooled to the least sensitive volume, since pooling would discard genuine quantified counts a more sensitive plating of the same sample recorded (Table S12 note).

A volume-derived floor is a point mass, with nothing to infer. A floor inferred from a pile-up on a most-probable-number rung carries a discrete posterior over the rungs at or below the observed minimum; the interval quoted is the 95 per cent highest-posterior-density support. Where no floor is evidenced, or the support spans more than one $\log_{10}$, observability labels are **refused** rather than reported (Table S15): the clinical deposit passes this rule at 0.40 $\log_{10}$ of support, and two of the five deposits fail it. This paper fits nothing to predict an MDK: once L is in hand the rest is derived.

The same likelihood turns classification into an inference with its own refusal rule: given a floor reading, the posterior over the true count maps through starting density to a posterior over class, and an isolate is refused when no class reaches 95 per cent confidence (Table S11). The refusal holds across both well designs, both floor readings and two priors.

Reachability and identifiability boundaries

For a culture starting at N_0 against an assay floor L , the **headroom** is $h = \log_{10}(N_0/L)$, the deepest reduction the assay can resolve. Two boundaries follow from the definitions alone.

Reachability. A q -log endpoint is observable only if

$$N_0 \geq N_{\text{reach}} = L \cdot 10^q.$$

Identifiability. A censored reading places the true count in $[0, L]$, so the recorded surviving fraction is at most L/N_0 (an upper bound, not a measurement). Given class thresholds $0 < c_1 < c_2 \dots$, the class is determined by the data only if

$$N_0 > N_{\text{id}} = L/c_1,$$

since the censored interval always contains zero and so is only compatible with the lowest class above this point. Both boundaries are properties of the method alone; seeding a culture above $L \cdot \max(10^q, 1/c_1)$ makes both failure modes impossible before an experiment runs.

Statistical analysis

$\log_{10}$ viable counts were modelled as a linear function of time using a **Tobit model** with observation-specific left-censoring limits, fitted by maximum likelihood (29); a censored reading contributes $\log \Phi((\text{limit} - \mu)/\sigma)$,

the probability it fell below its own limit (Beal's M3 written for this assay (30)). As a computational consistency check, 50 imputed datasets were generated from the fitted truncated-normal distribution, analysed by ordinary least squares, and pooled using Rubin's rules (31); both procedures share the same distributional assumptions, and their agreement shows the answer does not depend on which arithmetic recovers it, not that the sub-floor distribution is normal.

The growth proxy is the deposit's time to an optical density of 0.4, in days, so it runs inversely to growth rate. **The tolerance label** was modelled as an ordered categorical outcome by proportional-odds ordinal logistic regression (32), with starting density as a prespecified covariate; proportional odds was tested by a Brant test per predictor (33) and a likelihood-ratio test against a generalised ordered logit (34), with the partial-proportional-odds fit reported where it fails and checked against a multinomial fit assuming no ordering (Table S16).

Time-to-event analysis treated the first observed crossing below the assay floor as the event, right-censoring series that never fall below it; 60 per cent of series that cross read above the floor again at a later visit, so the event is a crossing, not a clearance. The crossing time is interval-censored between the last visit above the floor and the first below it; interval-censored fits (35) are reported alongside Kaplan–Meier (36) and Cox proportional hazards (37) as descriptive summaries, with cluster bootstraps over laboratories and flasks (38) in place of the partial likelihood's independence assumption, and proportionality tested on Schoenfeld residuals (39, 40).

The replicate-level bootstrap (41) behind interval slopes resamples the three replicate counts with replacement, takes their mean, refits the concentration slope on each draw, and takes percentile intervals from 20,000 draws, reported instead of the least-squares fit through the deposited concentration means, which has two residual degrees of freedom and discards the replicate scatter. An interval on a proportion is the Jeffreys interval (42), which does not collapse to zero width at the sample sizes here.

Causal mediation of the resistance association. A linear product-of-coefficients mediation decomposes the total effect of isoniazid resistance on the day-5 tolerance class into an average causal mediation effect through $\log_{10} No$ and an average direct effect (43, 44), with bootstrap percentile intervals from 5,000 resamples. A refit excludes the 18 of 203 isolates whose day-5 reading sits at or below the floor, since their recorded fraction carries no measured numerator. Sequential ignorability is assumed and untestable here; the sensitivity that matters is the residual correlation between mediator and outcome errors at which the point estimate crosses zero, about -0.23 . No claim is made that unmeasured confounding is absent.

Denominators. A dozen denominators appear in this paper and each is derived in a single place (Table S3): the MDR tolerance label is dropped wherever an ordered outcome is fitted (14 rows per panel, 20 at 60 days), the growth proxy is missing for one isolate, and five of 72 treated flasks in the six-laboratory deposit carry no usable starting density. Dropped rows are compared with retained ones on starting density and susceptibility to test whether the exclusions are plausibly ignorable (Table S17).

Decomposition of a crossing time. Over an interval where the decline is close to log-linear, $\log_{10} N(t) = a - bt$, and with ℓ the log₁₀ assay floor and $D = a - \ell$ the distance the population starts above it, the crossing time is $T = D/b$. For two flasks,

$$\log(T_A / T_B) = \log(D_A / D_B) - \log(b_A / b_B),$$

which splits a difference in crossing time into a distance term and a rate term. An **inversion** is a pair in which A fell faster yet crossed later, occurring exactly when $D_A/D_B > b_A/b_B$.

Multiplicity. Where a question admits more than one test, every test the deposit supports is run, and the family is corrected by the Benjamini–Hochberg procedure (45) at a false discovery rate of 5 per cent.

Four procedures are implemented in the released analysis code rather than taken from a package: the Brant test, the partial-proportional-odds fit, the product-of-coefficients mediation, and the residual-correlation sensitivity. These live in `exp30_ordinal_tolerance.py` and `exp34_mediation.py`; the partial-proportional-odds likelihood is validated against the proportional-odds fit it nests, agreeing to 1.0×10^{-10} in log-likelihood.

Analyses used Python 3.14 with numpy 2.5.0 (46), scipy 1.18.0 (47), pandas 3.0.3 (48), statsmodels 0.15.0 (49) and lifelines 0.30.3 (50).

Reproducibility

Every number in this paper is regenerated by a script that writes a machine-readable receipt, and an audit script recomputes 174 quantities quoted in the text from the tables they came from. A second audit checks that every number in the Abstract, Box 1, and every table and figure legend traces to a results file or receipt, that no numeric literal carries two incompatible statistical labels (a hazard ratio quoted where another table calls the same figure a p-value, for example), and that no cross-reference or sample-size count is stale. Neither audit pins every cell of every table.

Ethics

This study analysed publicly available, de-identified datasets, and generated one time-kill experiment in a reference strain of *Escherichia coli*. No human or animal subjects were involved, no patient samples were collected, and no individual is identifiable from any material presented here. The prospective experiment was performed at the Department of Microbiology, Iran University of Medical Sciences, Tehran, Iran, on a biosafety-level-1-appropriate quality-control strain (ATCC 25922); no institutional review board holds jurisdiction over secondary analysis of de-identified public data or over work on a bacterial quality-control strain.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Anthropic's Claude was used under the author's direction to read the manuscript and to assist in running the Python analysis pipeline. Claude Sonnet 5 was further used, under the author's direction, to document and organise the analysis code and to prepare its commits to the version-controlled repository. No figure, image or schematic was generated by an artificial intelligence tool; every figure is plotted from values that regenerate from the public deposits. No artificial intelligence tool is an author. The author directed the study, verified every reported quantity against the source deposits and the regenerating pipeline, and takes full responsibility for the content of this article, including its accuracy and its originality.

Data availability

Every dataset analysed here was already public under an open licence. Each is identified in Table 3 and cited alongside the article that first described it: Vijay and colleagues (21, 51), van Wijk and colleagues (8, 9), Windels and colleagues (23, 24), Kaur and colleagues (25, 26), and the hollow-fibre Source Data of Dubey and colleagues (10, 11), held out of every fitting step. The readings from the prospective experiment are deposited with the analysis code.

That code, both audit scripts, the headroom tool and machine-readable receipts recording the software versions each stage ran under are in a public repository (52), dual-licensed: the code under the Massachusetts Institute of Technology (MIT) licence, and the manuscript, figures and results under CC BY 4.0. The deposits keep their depositors' own licences. It is at <https://github.com/piranfar/detection-floor-tolerance-endpoints>, and this manuscript was built from commit c00066d+ of it.

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The author declares no competing interests.

Vahhab Piranfar is the sole author and is responsible for conceptualisation, methodology, software, formal analysis, investigation, data curation, writing of the original draft, review and editing, and visualisation.

Figure legends

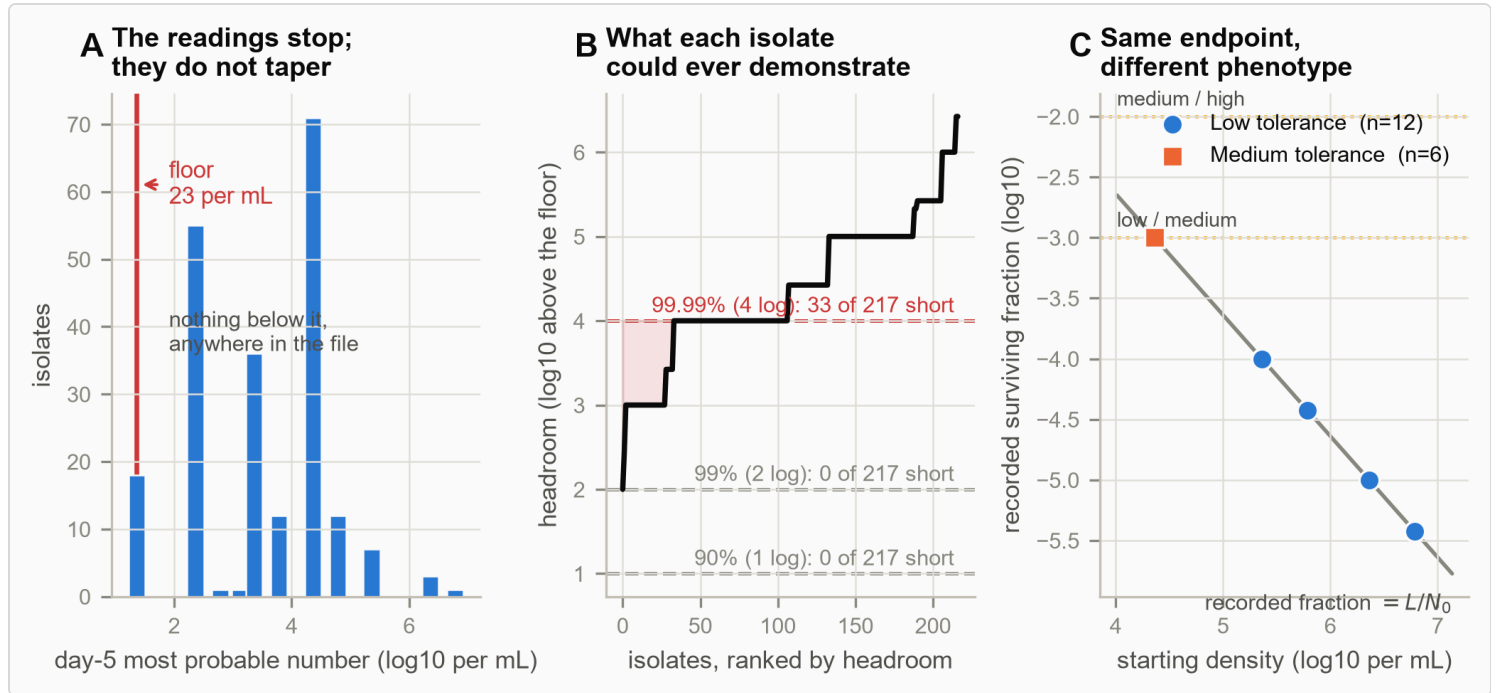


Figure 1. A log-reduction endpoint is bounded by the assay floor, and the bound decides the phenotype. 217 clinical *M. tuberculosis* isolates under rifampicin, from the deposit that carries the tolerance classification. **(A)** The distribution of day-5 most probable numbers. It stops at 23 per mL with a pile-up on it rather than tapering, and no value anywhere in the file lies below; that is the behaviour of a floor. **(B)** Headroom, the distance from each isolate's starting density down to that floor, ranked across isolates, against the depth each tolerance endpoint requires. No isolate is short of the 90 or 99 per cent endpoint. Thirty-three of 217 fall in the shaded band below the 99.99 per cent endpoint, and all 33 are recorded as failing to reach it. **(C)** The eighteen isolates of the 15-day panel whose day-5 reading was censored at the floor. Their true final counts are unknown below L , so the assay cannot distinguish their final viable burdens; what is plotted is the recorded fraction L/N_0 , which is what a censored reading computes, so the 265-fold span in apparent survival is exactly the 265-fold span in starting density. The dotted lines are the deposited classification thresholds. Which side of the low/medium cut an isolate falls on is settled by where it began, not by what the reading measured, though reaching the floor at all required rifampicin to cover that isolate's whole headroom, at least 3.00 logs for the shallowest and 5.42 for the deepest. For the twelve below the cut, no count the assay admits leaves any class but **low** compatible; for the six that sit exactly on it the compatible range straddles the cut and two classes remain.

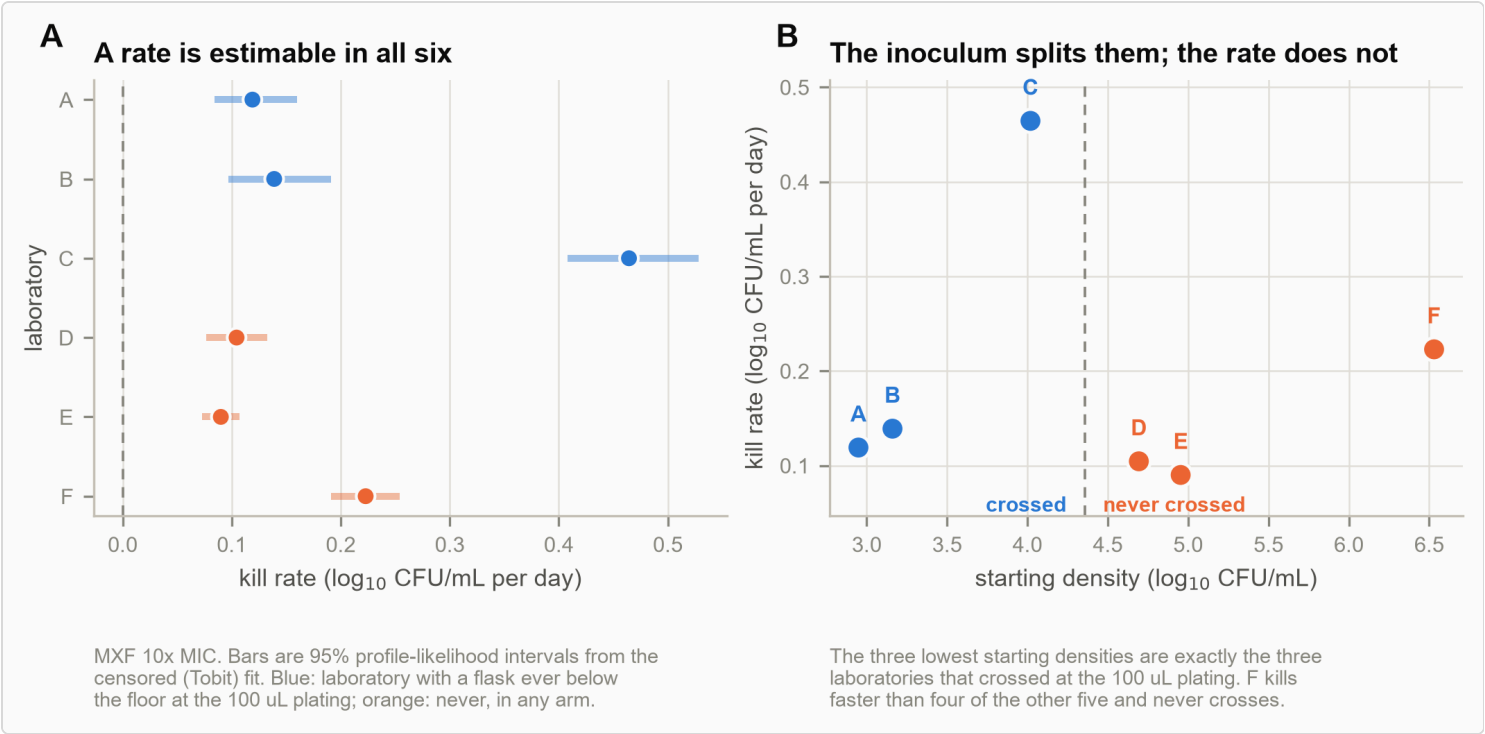


Figure 2. Where the inoculum is set by protocol, the two summaries still diverge. Six laboratories, one written protocol, one stock of *M. tuberculosis* H37Rv, moxifloxacin at ten times the minimum inhibitory concentration. **(A)** The kill rate by censored maximum likelihood with 95 per cent profile-likelihood intervals; every laboratory yields one, spanning 5.2-fold. **(B)** The two summaries against each other. The three lowest starting densities are exactly the three laboratories that recorded crossings at the 100 μ L plating; the kill rate produces no such separation. The crossing times themselves are Figure S1, with the laboratory-level tests this design does and does not support.

Supplemental material

The supplemental file carries the full Materials and Methods, the analyses named above as supplementary text, and 26 supplemental tables. Of those, 9 belong to analyses reported in that file rather than in this article, and are cited there at the point where each is used.

Tables

Table 1. The reduction each tolerance endpoint requires against the reduction the assay can resolve.

Prior culture	Endpoint	Isolates	Short of headroom	Fraction short	At ceiling: short vs ample
15 days	90% (1 log)	217	0	0.0%	-
15 days	99% (2 log)	217	0	0.0%	-
15 days	99.99% (4 log)	217	33	15.2%	100% vs 88%
60 days	90% (1 log)	210	0	0.0%	-
60 days	99% (2 log)	210	0	0.0%	-
60 days	99.99% (4 log)	210	7	3.3%	100% vs 95%

Note. In the 15-day and the 60-day prior-culture panel alike. Headroom is the distance from an isolate's starting density down to the day-5 MPN floor of 23 per mL. An isolate short of headroom cannot reach that endpoint however completely the drug worked, and every such isolate is recorded at the assay ceiling.

Table 2. Isolates whose day-5 reading was censored at the MPN floor.

Prior culture	At the floor	Starting density spread	Recorded survival spread	Labels assigned at the floor	Ordered calls
15 days	18	265x	265x	Low: 12, Medium: 6	203
60 days	6	100x	100x	Low: 6	197

Note. Starting densities differ across the row, so the same floor-level reading implies a different range of compatible fractional reductions in each; the recorded fraction is L/No, so the spread in apparent survival equals the spread in starting density exactly, and the labels differ accordingly. The last four columns sort every call in the 15-day panel, not only the censored ones: a censored reading bounds the class from above, and sweeping the true count across the admissible range leaves twelve calls with a single compatible class and six with more than one. Reaching the floor at all is a property of the killing; which class is then compatible is a property of the starting density and the floor. Ordered calls is the denominator those four columns are taken over, and it is not the isolate count of Table 1: it counts only the isolates whose deposited label is one of the three ordered classes, leaving out the 14 at 15 days and the 13 at 60 days that carry the unordered fourth category.

Table 3. The five published deposits reanalysed.

Dataset	Organism	Drug and range	Design	Deposit	Licence
Six-laboratory exercise	<i>M. tuberculosis</i> H37Rv	moxifloxacin, isoniazid, 1x and 10x MIC	90 flasks, 2 775 readings	figshare 19766083	CC BY 4.0
Clinical isolates	<i>M. tuberculosis</i> , 217 isolates	rifampicin	6 duration endpoints per isolate	eLife 93243, suppl. file 2	CC BY 4.0
Evolved clones	<i>E. coli</i> , 126 clones	amikacin	MIC and persister fraction per clone	Zenodo 7550302 (23)	CC BY 4.0
Concentration-by-time grid	<i>M. tuberculosis</i>	apramycin 1-128 ug/mL (amikacin arm not analysed)	5 concentrations x 4 days x 3 replicates	figshare 26462791	CC BY 4.0
Analysed last, after the derivation was fixed	<i>E. coli</i> , hollow fibre	amoxicillin-clavulanate	20 cultures, measured day-zero density, 100 uL plated	Nat Commun 2026 Source Data	CC BY 4.0

Note. Four carry the analysis and the fifth was analysed last, to test it. None was generated for this study. That fifth deposit was opened after every boundary and threshold was fixed: the headroom tool at 00:08 and the symbolic recovery of both boundary laws at 01:34 on 6 September 2026, against the deposit's arrival at 02:41 the same morning. The deposit entered in the same commit as the script that tests it, which git cannot order internally, so it is where reachability and identifiability were committed that dates the hold-out, not where the deposit arrived. This is an attestation and not a registration: the deposit had been public since its article appeared on 13 June 2026, so nothing outside this repository's own commit history evidences that it was not opened earlier, and a reader should weigh it accordingly. The other four were selected for the fields they carry, and that selection is not separately registered.

References

1. World Health Organization. 2022. WHO consolidated guidelines on tuberculosis. Module 4: treatment — drug-susceptible tuberculosis treatment. World Health Organization, Geneva, Switzerland. <https://iris.who.int/handle/10665/353829>.
2. Brauner A, Fridman O, Gefen O, Balaban NQ. 2016. Distinguishing between resistance, tolerance and persistence to antibiotic treatment. *Nat Rev Microbiol* 14:320-330. <https://doi.org/10.1038/nrmicro.2016.34>.
3. Brauner A, Shores N, Fridman O, Balaban NQ. 2017. An experimental framework for quantifying bacterial tolerance. *Biophys J* 112:2664-2671. <https://doi.org/10.1016/j.bpj.2017.05.014>.
4. Balaban NQ, Helaine S, Lewis K, Ackermann M, Aldridge B, Andersson DI, Brynildsen MP, Bumann D, Camilli A, Collins JJ, Dehio C, Fortune S, Ghigo J-M, Hardt W-D, Harms A, Heinemann M, Hung DT,

- Jenal U, Levin BR, Michiels J, Storz G, Tan M-W, Tenson T, Van Melderen L, Zinkernagel A. 2019. Definitions and guidelines for research on antibiotic persistence. *Nat Rev Microbiol* 17:441-448. <https://doi.org/10.1038/s41579-019-0196-3>.
5. Cochran WG. 1950. Estimation of bacterial densities by means of the "most probable number". *Biometrics* 6:105. <https://doi.org/10.2307/3001491>.
 6. Blodgett R. 2023. BAM appendix 2: most probable number from serial dilutions. Bacteriological analytical manual. US Food and Drug Administration, Silver Spring, MD. <https://www.fda.gov/food/laboratory-methods-food/bam-appendix-2-most-probable-number-serial-dilutions>.
 7. Mann HB, Whitney DR. 1947. On a test of whether one of two random variables is stochastically larger than the other. *Ann Math Stat* 18:50-60. <https://doi.org/10.1214/aoms/1177730491>.
 8. van Wijk RC, Lucía A, Sudhakar PK, Sonnenkalb L, Gaudin C, Hoffmann E, Dremierre B, Aguilar-Ayala DA, Dal Molin M, Rybníček J, de Giorgi S, Cioetto-Mazzabò L, Segafreddo G, Manganelli R, Degiacomi G, Recchia D, Pasca MR, Simonsson USH, Ramón-García S. 2023. Implementing best practices on data generation and reporting of *Mycobacterium tuberculosis* in vitro assays within the ERA4TB consortium. *iScience* 26:106411. <https://doi.org/10.1016/j.isci.2023.106411>.
 9. van Wijk RC, Lucía Quintana A, Ramón-García S. 2023. *Mycobacterium tuberculosis* time kill assay data of the standardized protocol within the ERA4TB consortium. figshare. <https://doi.org/10.6084/m9.figshare.19766083.v1>.
 10. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*. *Nat Commun* 17:7504. <https://doi.org/10.1038/s41467-026-74323-2>.
 11. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Source Data to "Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*" (41467_2026_74323_MOESM4_ESM.xlsx). Nature Communications. <https://doi.org/10.1038/s41467-026-74323-2>.
 12. Evangelopoulos D, Prosser G, Rodgers A, Dagg B, Khatri B, Bhagwat A, Gonzalo X, Kolyva A, Bertozzi G, Silva-Pereira TT, Gibbons N, Bhatt A, Sabharwal N, Perdígao J, Portugal I, Rodrigues C, Duarte R, Gomes M, Cirillo DM, McHugh TD. 2022. Data from: Comparative evaluation of viable count, molecular bacterial load and most probable number for the enumeration of *Mycobacterium tuberculosis* in murine tissue. University College London Research Data Repository. <https://doi.org/10.5522/04/19175153.v2>. Retrieved 2026-09-07.

13. Mukamolova GV, Turapov O, Malkin J, Woltmann G, Barer MR. 2010. Resuscitation-promoting factors reveal an occult population of tubercle bacilli in sputum. *Am J Respir Crit Care Med* 181:174-180. <https://doi.org/10.1164/rccm.200905-0661OC>.
14. Chengalroyen MD, Beukes GM, Gordhan BG, Streicher EM, Churchyard G, Hafner R, Warren R, Otjombe K, Martinson N, Kana BD. 2016. Detection and quantification of differentially culturable tubercle bacteria in sputum from patients with tuberculosis. *Am J Respir Crit Care Med* 194:1532-1540. <https://doi.org/10.1164/rccm.201604-0769OC>.
15. Saito K, Warriar T, Somersan-Karakaya S, Kaminski L, Mi J, Jiang X, Park S, Shigyo K, Gold B, Roberts J, Weber E, Jacobs WR Jr, Nathan CF. 2017. Rifamycin action on RNA polymerase in antibiotic-tolerant *Mycobacterium tuberculosis* results in differentially detectable populations. *Proc Natl Acad Sci U S A* 114:E4832-E4840. <https://doi.org/10.1073/pnas.1705385114>.
16. National Committee for Clinical Laboratory Standards. 1999. Methods for determining bactericidal activity of antimicrobial agents; approved guideline. NCCLS document M26-A. National Committee for Clinical Laboratory Standards, Wayne, PA. ISBN 1-56238-384-1.
17. Vilchèze C, Jacobs WR Jr. 2007. The mechanism of isoniazid killing: clarity through the scope of genetics. *Annu Rev Microbiol* 61:35-50. <https://doi.org/10.1146/annurev.micro.61.111606.122346>.
18. Campbell EA, Korzheva N, Mustaev A, Murakami K, Nair S, Goldfarb A, Darst SA. 2001. Structural mechanism for rifampicin inhibition of bacterial RNA polymerase. *Cell* 104:901-912. [https://doi.org/10.1016/S0092-8674\(01\)00286-0](https://doi.org/10.1016/S0092-8674(01)00286-0).
19. March VFA, Mchedlishvili K, Goig GA, Maghradze N, Avaliani T, Aspindzelashvili R, Avaliani Z, Kipiani M, Tukvadze N, Jugheli L, Bouaouina S, Doetsch A, Kalkan S, Reinhardt M, Gagneux S, Borrell S. 2025. Within-host evolution of drug tolerance in *Mycobacterium tuberculosis*. *bioRxiv* 2025.07.29.667394; preprint, not peer reviewed. <https://doi.org/10.1101/2025.07.29.667394>.
20. ASTM International. 2023. E2315-23: Standard Guide for Assessment of Antimicrobial Activity Using a Time-Kill Procedure. ASTM International, West Conshohocken, PA. <https://doi.org/10.1520/E2315-23>.
21. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Supplementary file 2 to "Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study" (elife-93243-sup2-v1.xlsx). *eLife*. <https://doi.org/10.7554/eLife.93243>.
22. Rabodoarivelo MS, Hoffmann E, Gaudin C, Aguilar-Ayala DA, Galizia J, Sonnenkalb L, Dal Molin M, Cioetto-Mazzabò L, Degiacomi G, Recchia D, Rybniker J, Manganelli R, Pasca MR, Ramón-García S, Lucía A. 2025. Protocol to quantify bacterial burden in time-kill assays using colony-forming units and

most probable number readouts for *Mycobacterium tuberculosis*. STAR Protoc 6:103643.

<https://doi.org/10.1016/j.xpro.2025.103643>.

23. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence. ISME J 18:wrae070. <https://doi.org/10.1093/ismejo/wrae070>.
24. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Data supporting "Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence". Zenodo. <https://doi.org/10.5281/zenodo.7550302>.
25. Kaur P, Ramya VK, Naveenkumar CN, Bharathkumar K, Singh M, Hobbie SN, Shandil RK, Narayanan S. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis*. Front Trop Dis 5:1413211. <https://doi.org/10.3389/fitd.2024.1413211>.
26. Kaur P. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis* — raw data. figshare. <https://doi.org/10.6084/m9.figshare.26462791.v1>.
27. Clinical and Laboratory Standards Institute. 2024. Methods for dilution antimicrobial susceptibility tests for bacteria that grow aerobically, 12th ed. CLSI standard M07. Clinical and Laboratory Standards Institute, Wayne, PA.
28. Clinical and Laboratory Standards Institute. 2026. Performance standards for antimicrobial susceptibility testing, 36th ed. CLSI supplement M100. Clinical and Laboratory Standards Institute, Wayne, PA.
29. Tobin J. 1958. Estimation of relationships for limited dependent variables. Econometrica 26:24. <https://doi.org/10.2307/1907382>.
30. Beal SL. 2001. Ways to fit a PK model with some data below the quantification limit. J Pharmacokinet Pharmacodyn 28:481-504. <https://doi.org/10.1023/A:1012299115260>.
31. Rubin DB. 1987. Multiple imputation for nonresponse in surveys. John Wiley & Sons, New York, NY. <https://doi.org/10.1002/9780470316696>.
32. McCullagh P. 1980. Regression models for ordinal data. J R Stat Soc Series B Stat Methodol 42:109-127. <https://doi.org/10.1111/j.2517-6161.1980.tb01109.x>.
33. Brant R. 1990. Assessing proportionality in the proportional odds model for ordinal logistic regression. Biometrics 46:1171-1178.
34. Peterson B, Harrell FE Jr. 1990. Partial proportional odds models for ordinal response variables. J R Stat Soc Ser C Appl Stat 39:205. <https://doi.org/10.2307/2347760>.

35. Turnbull BW. 1976. The empirical distribution function with arbitrarily grouped, censored and truncated data. *J R Stat Soc Series B Stat Methodol* 38:290-295. <https://doi.org/10.1111/j.2517-6161.1976.tb01597.x>.
36. Kaplan EL, Meier P. 1958. Nonparametric estimation from incomplete observations. *J Am Stat Assoc* 53:457-481. <https://doi.org/10.1080/01621459.1958.10501452>.
37. Cox DR. 1972. Regression models and life-tables. *J R Stat Soc Series B Stat Methodol* 34:187-202. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>.
38. Field CA, Welsh AH. 2007. Bootstrapping clustered data. *J R Stat Soc Series B Stat Methodol* 69:369-390. <https://doi.org/10.1111/j.1467-9868.2007.00593.x>.
39. Schoenfeld D. 1982. Partial residuals for the proportional hazards regression model. *Biometrika* 69:239-241. <https://doi.org/10.1093/biomet/69.1.239>.
40. Grambsch PM, Therneau TM. 1994. Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika* 81:515-526. <https://doi.org/10.1093/biomet/81.3.515>.
41. Efron B. 1979. Bootstrap methods: another look at the jackknife. *Ann Stat* 7:1-26. <https://doi.org/10.1214/aos/1176344552>.
42. Brown LD, Cai TT, DasGupta A. 2001. Interval estimation for a binomial proportion. *Stat Sci* 16:101-133. <https://doi.org/10.1214/ss/1009213286>.
43. Imai K, Keele L, Yamamoto T. 2010. Identification, inference and sensitivity analysis for causal mediation effects. *Stat Sci* 25:51-71. <https://doi.org/10.1214/10-STS321>.
44. Baron RM, Kenny DA. 1986. The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations. *J Pers Soc Psychol* 51:1173-1182. <https://doi.org/10.1037/0022-3514.51.6.1173>.
45. Benjamini Y, Hochberg Y. 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc Series B Stat Methodol* 57:289-300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
46. Harris CR, Millman KJ, van der Walt SJ, Gommers R, Virtanen P, Cournapeau D, Wieser E, Taylor J, Berg S, Smith NJ, Kern R, Picus M, Hoyer S, van Kerkwijk MH, Brett M, Haldane A, del Río JF, Wiebe M, Peterson P, Gérard-Marchant P, Sheppard K, Reddy T, Weckesser W, Abbasi H, Gohlke C, Oliphant TE. 2020. Array programming with NumPy. *Nature* 585:357-362. <https://doi.org/10.1038/s41586-020-2649-2>.
47. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones

- E, Kern R, Larson E, Carey CJ, Polat İ, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 1.0 Contributors. 2020. SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods 17:261-272. <https://doi.org/10.1038/s41592-019-0686-2>.
48. McKinney W. 2010. Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference 56-61. <https://doi.org/10.25080/Majora-92bf1922-00a>.
49. Seabold S, Perktold J. 2010. Statsmodels: econometric and statistical modeling with Python. Proceedings of the 9th Python in Science Conference 92-96. <https://doi.org/10.25080/Majora-92bf1922-011>.
50. Davidson-Pilon C. 2019. lifelines: survival analysis in Python. J Open Source Softw 4:1317. <https://doi.org/10.21105/joss.01317>.
51. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study. Elife 13:RP93243. <https://doi.org/10.7554/eLife.93243>.
52. Piranfar V. 2026. Analysis code, audit scripts and headroom tool for "The detection floor bounds tolerance endpoints in Mycobacterium tuberculosis". GitHub. <https://github.com/piranfar/detection-floor-tolerance-endpoints>.

SUPPLEMENTAL MATERIAL

Supplemental material

Rifampicin tolerance classification is bounded by assay detection floor among clinical *Mycobacterium tuberculosis* isolates

Vahhab Piranfar

Supplementary text

Extended background for the Introduction

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The bound is already in the papers that defined the assays. Vijay and colleagues, introducing the most-probable-number MDK for rifampicin (a count read off a dilution series rather than off colonies on a plate) treated an MDK99.99 longer than ten days as high tolerance beyond the **time** window of the assay, not as a statement about the count floor (1). In the 217-isolate classification that follows from that assay they report that MDK99.99 at 15 days of recovery could be calculated for only 22 of 209 isolates and that the rest lay, in their words, "beyond the assay limits" (2). The ERA4TB consortium requires below- and above-quantification-limit flags together with the limit of quantification, and notes that pharmacometric models can use those flags (3); the same six-laboratory comparison then excluded both from its numerical analysis. Their 2025 protocol states the geometry of the pipette explicitly: a 2.5 μL drop has a limit of detection above 2.6 $\log_{10}$ CFU/mL, and a lower limit requires a larger plated volume (4). Independently, a 2025 pharmacokinetic-pharmacodynamic analysis of hollow-fibre CFU series showed that censoring counts below 10 CFU (recording them only as "below the floor", with the true value unknown) biases regimen ranking (5), and a within-host tuberculosis tolerance study chose a shallower MDK threshold specifically so that more isolates could enter the analysis (6).

The closest statement of the bound is in the framework's own methods paper, and we claim no priority over it. Brauner and colleagues, presenting a direct measurement of tolerance that deliberately avoids time-kill curves, scale the inoculum to the endpoint being measured: a hundred bacteria per well to determine MDK99, a thousand for MDK99.9, and so on (7). That is the reachability inequality of this paper, with the floor idealised at one cell per well and a grew-or-did-not-grow readout in place of a count. What has not been done is to carry the rule back into plate-count time-kill, where the floor is not one cell but a concentration fixed by the plated volume (10 to 400 CFU/mL across the deposits reanalysed here) and must therefore be measured sample by sample rather than assumed; nor to ask what the published record looks like where the rule was not applied.

The requirement is older still, and it is quantitative: NCCLS M26-A ties the plated volume to the endpoint, and separately requires the smallest accurately detectable count to be established by serial dilution of a known inoculum (8). The endpoint has since moved and the rule did not travel with it. The persistence field's own consensus guideline, which catalogues at length what else can go wrong with a time-kill assay, raises neither the limit of detection nor the plated volume nor how deep a reduction the assay can report (9).

This is testable rather than arguable, because one published deposit assigns the tolerance phenotype itself. Vijay and colleagues (2, 10) scored 217 clinical *Mycobacterium tuberculosis* isolates as having low, medium or high tolerance to rifampicin, and deposited alongside each call the starting most probable number, the growth rate,

the isoniazid susceptibility and the killing readings the call was built from. The classification, the inputs to it, and the assay geometry that bounds it are all in the same file.

We test the two boundaries on the classification and the assay geometry of 217 clinical isolates, then ask what the resulting phenotype tracks: drug susceptibility, or the physiological state of the culture. Four further deposits carry the question beyond one clinical panel: a six-laboratory consortium exercise distributing one strain under one written protocol (3, 11), a concentration-by-time grid in the same organism (12, 13), a panel of evolved clones in which the concentration and duration axes can be checked for independence (14), and a hollow-fibre deposit (15, 16) held out until every boundary was fixed. All five are published and openly licensed (Table 3). The two boundaries are then tested forward rather than backward, in a prospective experiment built to break them, with every prediction written down before a plate was counted.

Two of the claims this arithmetic supports are not testable and should not be read as though they were. If the last reading sits at the floor, the recorded reduction cannot exceed h , because a deeper one would require a count below L . And the recorded fraction is then L/N_0 , a monotone decreasing function of N_0 , so its rank correlation with N_0 is -1 whatever the numbers are, including random ones. Both follow from the definition of a censored reading, and neither is evidence about a drug.

What is testable is what happens when one sample is plated at two volumes under a fixed class threshold. If both readings sit at their own floors the labels are L_1/N_0 and L_2/N_0 , and they differ only when those two fractions fall on opposite sides of the cut, which is again arithmetic. The empirical quantity is how often a real experiment lands in that window. It could be none of the time.

Extended background for the Discussion

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The minimum duration for killing was built to divide the starting density out, and in the definition it does. It stops the moment killing reaches the plate's floor, for a reason that is arithmetic rather than statistical: from there the fraction written down is L/N_0 , and the drug has dropped out of it.

Fifteen per cent of the clinical isolates examined here could not have shown the deepest endpoint whatever the drug did to them, and every one of them is recorded as having failed to show it. Eighteen more ended on the floor, and one threshold on their starting density reproduces every label they were given. Whether those cultures reached the floor is a question about rifampicin; which label they received once they were there is a question about arithmetic, and the starting density and no further. In the 15-day panel, 12 of 203 usable calls have a single compatible class and 6 have more than one; the other 185 rest on a fraction that was actually measured. The kill rate still carries more of the variance in crossing time than the distance does, in every multi-laboratory arm. What fails is the *interpretability* of a deep log-reduction call reported without N_0 and L , not every tolerance call ever made.

The failure is specific and bounded. In this deposit neither the 90 nor the 99 per cent endpoint is compromised: every isolate has room to show both. It is the 99.99 per cent endpoint, the deep endpoint the clinical classification uses, that a substantial minority cannot reach. Being that specific is what makes the problem fixable rather than fatal.

This is a derivation rather than an observation, which is what lets anyone test it on their own plates. Both boundaries follow from the definitions alone and neither refers to the drug or the strain: give them a floor, a set of class thresholds and a starting density, and they name the affected isolates in advance. In the clinical deposit they name them exactly, and the part of that agreement which could have gone the other way, the six isolates sitting on N_{id} itself, where the strict inequality leaves the class undecidable, goes the way the boundary predicts. In a deposit the framework was not built from it locates the depth at which a sterilisation claim stops being demonstrable: five of twenty cultures cannot show a five-log reduction, and none of the twenty can show six.

Everything above is read backwards, out of files made for other purposes, and a derivation that only explains the past is worth less than one that survives being aimed at the future. N_{reach} and N_{id} were therefore used to design an experiment against themselves, predictions written down before a plate was counted. At the inoculum the standard specifies the four-log endpoint was unreportable at the 10 μ L plating and comfortable at 100 μ L, same flask, same afternoon, which is arithmetic and could not have failed. What could have failed is whether a real experiment ever lands in the window where the two platings fall on opposite sides of a class threshold, and at the cut the clinical classification uses it does: one culture, two labels, and nothing between them but the pipette and the error in measuring what went in. That is reachability working as a design constraint rather than an explanation after the fact.

That is also where the counts feeding regimen selection are taken. A model published as a preprint while this paper was in preparation predicts relapse in the murine model across nine datasets, 58 regimens and 2,239 relapse observations, reaching an area under the receiver operating characteristic curve of 0.90 on its final external validation (17). Its eight predictors include the fall in lung CFU from baseline at days 14 and 28, the fall in the ribosomal RNA synthesis ratio at day 14, the presence of an oxazolidinone, and four experimental covariates; removing either biomarker costs performance. The authors show the loss from dropping the RS ratio can be recovered by adding model-estimated coefficients for the sterilising contribution of each drug, a substitution they license only for compounds already characterised, expressly not for the novel ones the model exists to rank.

Their explanation for that gap is physiological, and it is why they built the RS ratio: a change in CFU cannot always separate a sterilising regimen from a non-sterilising one. A detection floor produces the same signature from outside. Where the best regimens drive the count into the region the plate cannot resolve, a term standing for drug identity absorbs discrimination the count is no longer free to express, and the fitted model looks as this one does. Nothing in the published record separates the two readings, because version 1 of that preprint states no detection limit and no plated fraction, and this is not a lapse peculiar to it, since the field's own consensus guideline raises neither. An explanation the reporting cannot adjudicate is one the reporting should not have

left open, and it is precisely why N_{reach} and N_{id} are worth writing down. A relapse model that leans on a 28-day count needs that count treated as left-censored, and h says when the question arises instead of leaving it to be noticed.

A reader may object that this runs backwards. Dropping below the limit of detection is what a *susceptible* population does; how can it make an isolate look more tolerant? The objection is fair, and it is answered by separating two endpoints the literature reports side by side.

A duration endpoint asks when a fixed depth was reached. An isolate without the room that depth requires cannot reach it under any drug effect whatever, so what the file records is not a short duration but no duration at all: the isolate is censored at the end of observation, the top of the scale and therefore the most tolerant value it can take. The isolates that could not demonstrate the endpoint are, without exception, the ones recorded as having failed to reach it.

A class endpoint asks how far the population fell by a fixed time, and there the arithmetic does not merely preserve the direction, it inverts it. Once the final reading sits on the floor the recorded fraction is L/N_0 , and with L fixed a *lower* starting density yields a *larger* recorded fraction. A larger surviving fraction is a higher tolerance class, not a lower one. That is why the eighteen isolates ending on the same floor value split twelve low and six medium in the order their inocula do: the recorded survival and the starting density are the same quantity.

Neither route asks anyone to mistake a sterile culture for a surviving one. In the first the assay returns no endpoint; in the second it returns the floor divided by the inoculum. What is read as biology is, in both cases, a property of the measurement.

The same arithmetic answers the objection that the starting density is fixed by protocol. It is fixed as a turbidity, and a turbidity is not a viable count. Under one written protocol the densities actually achieved differed enough to move the deepest demonstrable kill by $2.33 \log_{10}$ between laboratories at a single plating volume, and the spread reaches $4.40 \log_{10}$ once laboratories and plated volumes are taken together; those two figures are ranges over different sets of readings, twelve at the $100 \mu\text{L}$ plating against fifty-six across every laboratory and volume, and ranges do not add. What is exact is the volume term on its own: floors of 10, 100 and 400 CFU/mL , set by plating 100, 10 and $2.5 \mu\text{L}$, span $1.60 \log_{10}$ by arithmetic on $L = 1000/v$, with no sampling uncertainty at all. The density term carries all of the uncertainty, a range over the four laboratories that deposit an untreated day-zero reading at the $100 \mu\text{L}$ plating, and resampling whole laboratories runs from 0.05 to $2.33 \log_{10}$, bounding that term from below and not from both sides; it could be almost nothing. The exact part is the stronger half, and the choice of plated volume is what sets the floor.

The isoniazid-resistance association behaves differently, and instructively. It is the weaker of the two on two separate counts. Unadjusted, resistant isolates carry 2.32 times the odds of a higher tolerance class; adjusted for starting density the odds ratio is 1.31 and the interval spans one. It also does not clear its corrected threshold once the analysis is restricted to the baseline isolates, one per patient by construction, the only correction for

repeated isolates this deposit permits, since it carries no patient identifier. The growth association survives both.

Where the isoniazid-resistance association weakens, the weakening is visible in the same file: resistant isolates enter the assay ten-fold lower, and 26.2 per cent of them lack the room the endpoint requires. Why they seed lower is not known. The natural explanation, that resistance carries a fitness cost, is not supported here, since these isolates do not grow measurably more slowly and most carry the near-neutral *katG* S315X allele (18, 19). That gap is a finding in its own right and belongs in the next study rather than a speculative sentence in this one. Nor is it explained away by anything else the file records: of its nine pretreatment covariates only two can legitimately be adjusted for, and neither moves the association by more than five per cent (Section 4). The two larger movements in that table both come from columns that record the exposure rather than a confounder: the mutation identity, missing for all but one susceptible isolate, and the isoniazid MIC, which separates the two groups completely and enlarges the seeding gap by 13 per cent instead of shrinking it.

Where the inoculum is set deliberately rather than by clinical accident, the same arithmetic operates and can be measured rather than inferred. In more than a third of cross-laboratory comparisons the flask in which the population fell faster was the one that crossed below the assay floor later, and the distance-over-rate criterion accounts for three in five of them. Resampling whole laboratories rather than pairs leaves that rate poorly determined (3.0 to 72.4 per cent), so what the deposit establishes is the mechanism and not the rate.

We agree with van Wijk and colleagues wherever the two analyses overlap. They noticed the spread in starting densities themselves, and report that the burden at the start varied between laboratories while the net effect of the drug varied less (3); that conclusion is theirs, and it anticipates part of ours. Where we part company is in what they did with readings that fell outside the quantification limits: they left them out of the numbers, a defensible way to report an assay, but one that closes off the question this paper asks, because those are exactly the readings a duration is built from.

Vijay and colleagues report that tolerance goes with resistance status and with treatment history in the same isolates (2). We do not dispute a single measurement; their deposit is unusually complete, and this analysis was only possible because they posted the classification, the readings behind it and the assay geometry together. What we add is this: their deepest endpoint asks for the largest fall, so it is the one shortest of room to fall in, and how much room there is differs systematically between the groups being compared. The link to resistance survives neither putting starting density into the model nor cutting the data down to one isolate per patient; the link to growth survives both. The finding is that one of its two associations is carried by the geometry of the assay and by isolates that are not independent of each other, not that their classification is empty.

The evidence between laboratories rests on six laboratories, not on hundreds of flasks. Every flask in a laboratory was seeded from the same starting culture, so for this purpose those flasks are copies of one another; the effective sample size for any comparison between laboratories is six, not the 67 to 85 flasks the tables list. Six is too few for a cluster-robust standard error, and the correction does not fail loudly when fitted anyway: on these data it returns a *smaller* p for every institute term than the model-based p it was meant to correct. So

where a comparison is between laboratories we quote the exact laboratory-level test or a bootstrap that resamples whole laboratories, and where the design cannot produce a meaningful p at all, we give none. Three laboratories recorded crossings at the 100 μ L plating and three did not; with the six split three and three, the smallest two-sided p the arithmetic can return is 0.10, whatever the biology. Some quantities do let flasks be pooled across laboratories, the variance shares and the inversion rate under a flask bootstrap: those keep an exact permutation test, and are the strongest between-laboratory evidence the deposit holds.

The laboratories did not all count their starting density on the same day. Only institutes C and D deposit a day-zero reading for the treated flasks; A, B, E and F deposit day one, by which point the ten-times-MIC arms have already lost between 0.7 and 2.4 $\log_{10}$. For four of the six laboratories, then, the number we are calling the exposure has some killing baked into it. Re-running the whole comparison on a day-one exposure for every laboratory left the ordering, the area under the curve and the laboratory-level p unchanged, so the conclusion is robust to it; the mismatch is real all the same, and is reported rather than smoothed over.

In the six-laboratory exercise, starting density and laboratory are close to the same variable. 87.0 per cent of the variance in flask-level starting density lies between laboratories rather than within them. Taken over every flask that deposits an early density, treated and untreated arms together, the laboratory means span 3.62 $\log_{10}$, against a median within-laboratory standard deviation of 0.43. Adding starting density to a model that already knows which laboratory a flask came from is therefore closer to swapping a label for a number carrying much the same information than to separating two covariates. Telling the two apart needs a design that fixes the inoculum across laboratories.

Starting density could sit on the causal path rather than beside it. Suppose resistance slows growth, and slow growth causes genuine tolerance: then the thin inoculum is a consequence of the slow growth, and adjusting for it would remove real signal rather than a confounder. The asymmetry favours the confounding reading, since growth survives adjustment and resistance does not, but a design that fixes the inoculum would settle it.

This is not an aspiration. One study already does it. A murine preventive-therapy experiment writes its lower limit of detection into the deposited workbook once on every sheet, separately for each agar type, and separately for individual mice where the lower limit differed between them: 0.78 $\log_{10}$ CFU per lung on plain and hygromycin agar, 0.54 on thiophenecarboxylic acid hydrazide agar, with the plated volume stated beside it (20). That is better practice than anything else in this corpus. The article reporting the experiment states none of it: a full-text search returns no occurrence of "limit of detection" or of either value. The number is not missing from the science; it is missing from the paper, and a reader who never opens the supplementary spreadsheet cannot know an endpoint was bounded. What we are asking for is one sentence in a methods section, and somebody has already done the work that sits behind it.

The mirror image is a deposit whose modelling table carries a column named `CFU_LOD`, empty in all 272 rows (21); nor does a detection limit appear in any of the eight analysis scripts deposited beside it. Put the two cases together and the diagnosis is not carelessness. The assay floor L is measured, sometimes to a standard higher

than anyone asks for, and then falls out of the record between the bench and the reader: off the end of a spreadsheet, or into a column nobody completed.

Model an ordered label as ordered. A tolerance class is low, medium or high. Scoring that 0, 1, 2 and fitting a line asserts that the step from low to medium is the same size as the step from medium to high, and no one has established that. Proportional-odds regression costs nothing to run and reports in odds ratios; where the proportionality it assumes fails, as it does here at 60 days, it says so rather than averaging two different effects into one null.

None of this requires new apparatus or a statistician, and to make that concrete the four checks are released as a small command-line tool alongside the analysis code. Give it a starting density and either an assay floor or a plated volume. It returns the headroom, states which of the 90, 99, 99.9 and 99.99 per cent endpoints that headroom can support, and reports the recorded fraction a floor-level reading would produce, labelled as the bound it is. Where a culture volume is known it also returns the viable burden a blank reading is consistent with; where it is not, it declines to compute one rather than assuming a volume. The tool exits with a failure code when a requested endpoint is unreachable, so it can be run before an experiment rather than after it.

Text S1. The turbidity standard fixes what goes into the flask, not what the plate can see

The inoculum for a time-kill experiment is not a free choice. A suspension is matched to 0.5 McFarland and diluted, and NCCLS M26-A (approved guideline, 1999; the body is now CLSI, which has archived the document) puts the target near 5×10^5 CFU/mL (8). If the protocol fixes the starting density, the variable the preceding sections rest on does not vary.

Two things let counts settle this instead. First, a McFarland reading is a turbidity (22), and cloudiness counts live cells, dead cells, debris and a clump of cells all as one particle; converting it to CFU/mL assumes a cell size, shape and dispersal that *M. tuberculosis* does not oblige, and tuberculosis protocols work from a range of dilutions and target inocula (23). Second, the standard fixes what goes into the flask, while what matters here is what the plate counts, and between the two sit a dilution, a transfer, and whatever clumping happens on the way.

In the six-laboratory exercise, the untreated flasks at day zero on the 100 μ L plating actually started at 3.67, 4.61, 4.65 and 6.00 log₁₀ CFU/mL, for institutes B, C, D and F. Institute E deposits no untreated day-zero reading at any volume; its series begins at day one. Institute A's three readings at that plating are flagged above the quantification limit rather than counted. Three of the four sit below the M26-A target, by 11-, 12- and 107-fold; the fourth sits two-fold above it. Every figure in this section is on that one basis: untreated, day zero, 100 μ L, which is not the basis of Table S5.

Read as headroom at a fixed plating volume, the spread is $\Delta h = 2.33$ log₁₀, a 216-fold difference in how far down the assay can look, under one written protocol (Table S18): B at $h = 2.67$, C at 3.61, D at 3.65, F at 5.00,

each sitting exactly one $\log_{10}$ below the density above it because $L = 10$ CFU/mL at this plating. Institute A's other platings put it near $h = 2.4$, so including it would widen the spread rather than narrow it. A range over four draws is a fragile statistic: its cluster-bootstrap 95 per cent interval runs from 0.05 to 2.33 $\log_{10}$, the upper limit being the observed range by construction.

The stronger version of the argument does not depend on that roster at all. Let the plated volume vary too and the spread widens to $\Delta h = 4.40 \log_{10}$, of which 1.60 $\log_{10}$ is exact arithmetic on $L = 1000/V$, carrying no sampling uncertainty whatever. The pipette moves the measurable depth further than the laboratories differ.

The obvious counter-explanation is early killing: if the drug had already acted by the day-zero reading, a spread in those readings would be kill and not inoculum. Only institutes C and D deposit a day-zero reading in both arms, and there the treated and untreated flasks agree to 0.06 and 0.13 $\log_{10}$, so for those two the spread is the inoculum and not early kill. The other four cannot be checked this way. Institute F shows why the check matters and why it is not decisive: its treated day-one readings average 6.62 $\log_{10}$, above its untreated day-zero reading of 6.00, but the untreated culture itself grows to 6.72 over the same twenty-four hours. A second deposit (15, 16) gives the same result against its own stated figure rather than a standard: its Methods specify 10^5 CFU/mL and its file measures 7×10^5 to 2.8×10^6 , a median of 1.215×10^6 and so 12.15-fold above nominal, computed from the unrounded median, which Table S18 displays rounded to 6.08 $\log_{10}$.

Distance from the McFarland reference is descriptive and reported as such (Table S19); sitting far below it is the intended state, since the inoculum is prepared by diluting from it, not a measure of protocol compliance. What sets the deepest log-kill an experiment can show is the measured starting density and the real assay floor, not the turbidity the preparation began from.

Text S2. The flask that fell faster often crossed the floor later

Take two flasks in the same treatment arm from different laboratories. Across 191 such pairs (45 further pairs the censoring could not settle are excluded rather than imputed), **66 pairs (34.6 per cent) are inversions**: the flask in which the population fell faster crossed below the assay floor later (Table S20). Restrict to pairs whose rates differ by more than 0.10 $\log_{10}$ per day, so the faster flask is unambiguously faster, and 23.0 per cent of 100 pairs still invert. The criterion $D_A/D_B > b_A/b_B$ calls 84.3 per cent of all 191 pairs correctly, but that figure is carried by the easy majority: 97.6 per cent of the 65.4 per cent that are not inversions, against 59.1 per cent of the inversions themselves. The decomposition identifies the mechanism, a pair inverts when the distance ratio exceeds the rate ratio, without predicting which pairs invert much better than three times in five; the residual is what a single averaged slope through a two-phase kill curve cannot capture, a known limitation of b stated in the Methods.

The effect holds when the faster flask is required to be unambiguously faster: raising the minimum separation between the two fitted rates from 0.02 to 0.05 to 0.10 $\log_{10}$ per day moves the inversion rate from 33.7 to 29.4 to 23.0 per cent.

Those point estimates are firm; their precision is not. The 191 pairs are built from 42 flasks in six laboratories, not independent coin flips. Resampling whole flasks within laboratory puts the 34.6 per cent between 22.3 and 48.7 per cent; resampling whole laboratories, the level at which the comparison is actually made, puts it between 3.0 and 71.3 per cent. Under the strictest rate separation the corresponding intervals are 9.4 to 42.9 per cent and 0 to 79.4 per cent. What the deposit supports is that inversions are common and are explained by the decomposition, not any particular rate.

The variance decomposition sets the size of the effect and bounds the claim. Of the variation available to move a crossing time, the distance term carries 35.2 per cent at ten times moxifloxacin, 32.3 per cent at ten times isoniazid and 17.7 per cent at one times isoniazid, with the rate term carrying the rest. **That ordering is not established, and we do not claim it.** The flasks are three to a laboratory and share a starting culture, so this is a between-laboratory statistic with six clusters wearing flasks as a disguise: deleting one laboratory pushes the distance share above a half in two of the three arms (70.4 per cent at ten times isoniazid, 62.3 per cent at ten times moxifloxacin), and a bootstrap over whole laboratories straddles a half in all three (Table S20). What survives is the weaker statement: the distance term is a large share of the variation available to move a crossing time, and the two terms are of comparable size.

The fourth treated arm, moxifloxacin at one times the inhibitory concentration, cannot be decomposed at all: five of six laboratories record net growth there, and only one flask returns a positive fitted rate, so there is no spread in the rate to divide (Table S20). The inoculum contributes a sizeable minority of the variance in crossing time, enough to flip which of two flasks fell faster in more than a third of the laboratory comparisons here, the practical stake for anyone choosing between compounds on this evidence.

Text S3. Two further deposits test how much drug against how long it takes, at a cost of reading late and pooling nutrient groups

The framework that defines these two measures takes it as given that they are separate (Brauner et al. 2016 (24)). Two secondary deposits bound how far from separate the two could be, without proving they are separate, which is weaker and more honest than a proof would be.

Apramycin against *M. tuberculosis* (12, 13) shows what reading a plate too late can do to a dose-response. Tested at 4 to 128 µg/mL (the 1 µg/mL arm grows rather than dies and is excluded), the survivor counts spread out 6.9-fold at day 3, 48.2-fold at day 7, and only 5.2-fold at day 14 (Fig. S3, Table S21): the spread opens, then closes again, and a 14-day readout would report this drug as nearly insensitive to a 32-fold change in dose. Each interval is fitted on its own, resampling all three replicate counts with replacement at both ends of every interval, 20,000 draws. Each doubling of the dose adds +0.059 log₁₀ per day to the kill rate over days 0-3 (95 per cent percentile interval +0.033 to +0.083), +0.033 over days 3-7 (+0.008 to +0.059), and -0.025 over days 7-14 (-0.035 to -0.015): concentration-dependence is positive early (early against late, +0.084, +0.055 to +0.111), smaller and not significant in the middle (early against middle, +0.026, -0.011 to +0.062), and reversed late. By day 14 the highest arm is down to 65 CFU/mL, the deposit states no limit of quantification and

no plated volume, so the late slope cannot be distinguished from a lower bound on a censored reading; what holds whatever the floor turns out to be is that the separation between doses collapses.

The 217-isolate clinical file, tested the other way, supports 24 comparisons of how much drug it takes to stop growth against how long it takes to kill, all corrected together (Benjamini-Hochberg against 24, Table S22). Four are significant before correction, where 1.2 are expected by chance, and none survives it; all four point the wrong way (a higher inhibitory concentration goes with a *shorter* killing time) and cluster at the deepest endpoint, the one Section 1 shows is compromised. Across all 24 tests the weakest correlation this design could have detected runs 0.14 to 0.25 depending on the subgroup; the six strongest, listed in Table S22, span 0.14 to 0.18. Restricted to baseline isolates alone, the deepest 15-day endpoint gives $\rho = -0.21$, $p = 0.0086$, short of the family threshold of 0.0021 by a wide margin, and short of 0.0083 by three and a half per cent even judged as its own family of six. A parallel test in 126 evolved *E. coli* clones sharing one ancestor (14, 25) returns $\rho = +0.043$, $p = 0.63$, against the 0.175 this sample size could have detected; independence holds inside every nutrient group separately, which matters because the concentration written on the flask is not the same exposure in every group (at 25 $\mu\text{g/mL}$, 0.55 to 9.47 times the inhibitory concentration that population had evolved to, Table S23). Neither deposit shows the two axes are independent. Both show they are not positively coupled, which is what they can support.

Extended results for article section 1: The plate's floor sets how far down killing can be seen

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The deposit never states its limit of quantification (10), so the floor has to be inferred, and the inference is quantified rather than asserted. A posterior over the MPN rungs at or below the lowest observed value places 95 per cent support on 9.2 to 23 per mL, a span of 0.40 $\log_{10}$, narrow enough that the class labels of Section 2 are computed rather than refused (Table S15). Where a deposit gives no such evidence, the labels are refused instead of reported.

The deposit carries a second classification, scored at day 2, on a different floor. The day-2 column bottoms out at 230 per mL, with eighteen readings resting there: $L = 230$, ten times the day-5 floor, a coincidence of this deposit rather than one number written into two places, since the two columns share no reading. Cuts at 10^{-2} and 10^{-1} , one decade shallower than at day 5, reproduce all 203 usable day-2 classes. The identifiability boundary does not move ($N_{\text{id}} = 230/10^{-2} = 23,000$ per mL, matching day 5), but the reachability boundary does, by a full decade: 121 of 203 isolates lack the room for a four-log reduction against the day-2 floor, against 31 against the day-5 floor. At 60 days no pair of decade cuts reproduces the day-2 classes.

Every $\log_{10}$ here is computed from the raw MPN rungs against a floor of 23 per mL and rounded once, at the end, so the deepest headroom displays as 6.42 rather than as the 6.43 that 7.79 minus 1.36 returns; the fold figures quoted throughout are likewise ten raised to the unrounded difference.

Two of those 33 carry a fourth label the deposit writes literally as "MDR" (conventionally multidrug-resistant), unordered with respect to the other three, so no ordered analysis can place it (Table S3); over the 203 isolates with an ordered label the count is 31.

No p-value is attached to that contrast: an isolate with less than four logs of headroom cannot record a four-log reduction, so the 0-of-33 cell is fixed by arithmetic and the null of independence between ceiling status and headroom is false before any data are seen. The census is the finding: every isolate that could not reach the endpoint is recorded as not having reached it, and 88.0 per cent of those that could are recorded the same way. The deepest endpoint is the only one at risk, and it is the one the tolerance classification is built on.

Extended results for article section 2: The same floor reading was given different tolerance labels

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The question this raises is which labels the rule could have returned, not whether those labels are wrong. Sweeping the true count across the censored interval $[0, L]$ and applying the deposited thresholds gives the set of classes compatible with each observation. For the twelve isolates that began at 230,000 per mL or above, every count the assay admits yields a fraction below 10^{-3} , so low is the only compatible class. For the six that began at 23,000, the compatible set spans the low-medium threshold and the rule cannot choose between two classes.

Three questions have to be kept apart, and the rest of this paper keeps them apart. Whether a culture reaches the floor at all is a question about the drug: rifampicin has to reduce the population by that isolate's entire headroom for it to happen. Which label the rule assigns once the reading is censored is a question about arithmetic: there the starting density determines the compatible set. How far the population actually fell is a question the assay leaves open.

So for twelve of the eighteen the recorded label carries no information beyond the fact that the reading was censored, and for six it is not determinate at all. None of the eighteen is a measurement of the surviving fraction. Sorting every call in the panel this way separates the labels the assay measured from the labels the censoring rule fixed (Table 2): at 15 days, 185 of 203 calls rest on a reading above the floor and are measured, 12 are censored with a single compatible class, 6 with more than one. At 60 days the counts are 191, 6 and none, because the cultures are denser and few readings reach the floor.

The shortfall falls unevenly on the groups a study would compare. The 99.99 per cent endpoint is unreachable for 22 of 84 isoniazid-resistant isolates, 26.2 per cent, against 9 of 119 susceptible ones, 7.6 per cent (Fisher exact $p = 0.00056$ (26)); the remaining two of the 33 carry the MDR label and fall outside that contrast. Median headroom differs by a full log, 4 against 5. A comparison of tolerance between those groups is therefore in part a comparison of how well each group could be measured.

Extended results for article section 3: The affected isolates are named before the drug is added

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

That agreement is an identity rather than a passed test. Once the label is a cut on the recorded fraction (Table S2) and every floor reading has a numerator of exactly L , the class of a floored isolate is a function of N_0 alone, so the algebra cannot disagree with the deposit. What could have failed, and did not, is the premise: that the three classes separate cleanly on the recorded fraction at all, in 203 of 203 calls.

How finely the agreement resolves has to be stated too. Among the eighteen the starting density takes only five values: 23,000 for six isolates, then 230,000 for eight, and 610,000, 2.3×10^6 and 6.1×10^6 for one, one and two, and **no isolate lies strictly between N_{id} and N_{reach}** . Any cut placed anywhere in that decade reproduces the same twelve-six split, so the split locates the boundary only to within an order of magnitude.

Two numbers in this section rest on reading the floor as a plate would, and both survive the assay's own likelihood: recomputing the twelve-six split under the MPN's own 95 per cent interval, reaching 120 per mL rather than stopping at 23, and under a one-sided censoring bound at 95 and 99 per cent, changes no verdict (Table S14); a posterior over class gives the same answer probabilistically, with the twelve keeping their label at posterior probability at least 0.999 and the six refused outright (Table S11).

Extended results for article section 4: The label tracks how fast an isolate grows, and its link to resistance cannot be separated from the inoculum

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The label has three levels in a fixed order, low, medium, high, and is modelled as such: proportional-odds ordinal logistic regression, which does not assume the step from low to medium equals the step from medium to high. That fit makes an assumption of its own, that a single odds ratio serves both cut points, and it was tested rather than assumed (Methods; Table S4).

The unadjusted 2.32 is the total effect of resistance on the label; the adjusted 1.31 is the controlled direct effect with the inoculum held fixed, under an assumption this deposit cannot test on its own, that starting density sits on the causal path as a mediator rather than beside it as a confounder. The reverse reading is equally available a priori: if slow growth both thins the inoculum and causes genuine tolerance, adjusting for starting density would remove real signal rather than a confounder (Discussion weighs the two readings against each other). What is not in question, whichever reading is right, is that the association weakens once starting density enters the model.

The total effect of resistance on tolerance class splits into a part that runs through $\log_{10}$ starting density and a part that does not. On the 203 isolates with a usable class, the mediated part is +0.166 classes (bootstrap 95 per cent CI +0.067 to +0.279), the direct part +0.091 (−0.116 to +0.287). About 65 per cent of the association travels through the inoculum; the direct path covers zero. Restricting to baseline isolates leaves the mediated path intact (+0.159, +0.053 to +0.295), and collapsing the outcome to two levels agrees (Table S24).

Two things bound that estimate. It rests on sequential ignorability, that nothing unmeasured drives both the starting density and the class, which no observational file can establish: a residual correlation of about −0.23 between the mediator and outcome errors would nullify the point estimate, and one ordinary measured covariate already moves it about forty per cent of that way. More fundamentally, the step in the middle is the denominator of the ratio the outcome is cut on: the tolerance class is a threshold on N_5/N_0 , so a decomposition that routes resistance through N_0 is in part recovering the censoring arithmetic this paper is about rather than a biological pathway from resistance to tolerance. That is not a reason to discard the estimate, since it says how much of the association is carried by the label's own denominator, but it is a reason not to read it as a mechanism. Only a design that fixes the inoculum separates the two.

That objection can be tested, not merely conceded, because the isolates it applies to can be named. The recorded fraction is L/N_0 exactly for the eighteen isolates whose day-5 reading sits on the floor, and for no others: in the remaining 185 the numerator was counted and is free to move independently of N_0 . Refit on those 185 alone and the mediated path does not weaken but strengthens, to +0.227 classes (+0.118 to +0.348), while the direct path collapses to +0.008 (−0.197 to +0.208) and the proportion mediated rises from 0.65 to 0.96. Deleting exactly the isolates the criticism is about leaves the effect larger, so it is not an artefact of them.

These 217 isolates are not 217 independent observations. Forty-three are follow-up isolates taken during treatment from patients who also gave a baseline isolate, so up to 86 rows sit in clusters of two. The deposit carries no patient identifier: Archive and Sample-ID are unique to each row, and none of the 48 columns repeats the way a patient key would have to, so the pairing cannot be recovered and no standard error can be clustered on the true grouping. What is available is the exact substitute, a baseline-only stratum in which every isolate is from a different patient by construction (Time_point = 0M, n = 167 at 15 days).

Why resistant isolates seed lower is not established here, and the obvious explanation fails: they do not grow measurably more slowly (median time to OD 0.4 of 19 against 17, $p = 0.24$), and 85 of them carry *katG* S315X, the mutation that predominates clinically precisely because it is close to fitness-neutral (18, 19). Seven of the nine pretreatment covariates the file carries cannot adjust this association, and they fail in two different ways. Five are recorded almost only for resistant isolates, so their missingness is the exposure: the two susceptibility calls are present for 82 of 84 resistant isolates and none of the 119 susceptible, the two Mykrobe calls for 76 and none, the mutation identity for 79 and one. Adjusting for such a column adjusts for resistance itself, which is why the two susceptibility rows agree to three significant figures: they are missing on exactly the same rows. The mutation identity's apparent 22 per cent attenuation is an artefact of that same circularity, not a real effect. The two Mykrobe columns hold a single value wherever they are recorded in this stratum, so no adjusted coefficient exists for them and Table S25 prints none.

The other two, the isoniazid and rifampicin MICs, are missing in the opposite direction: recorded for all 119 susceptible isolates and 67 of 84 resistant ones, so what is absent sits inside the exposed group rather than across the contrast. That does not make them confounders. The isoniazid MIC separates the two groups completely in this deposit, no resistant isolate overlaps any susceptible one, so it is the exposure measured on a graded scale rather than something to adjust the exposure for: the standard error on the resistance coefficient inflates from 0.103 to 0.181, the complete case drops 17 resistant isolates and no susceptible one, leaving 186 rows against 203, and with it in the model the seeding gap grows from -0.85 to -0.96 , an amplification of 13 per cent rather than an attenuation. The rifampicin MIC, which does not separate the groups, moves it by less than half a per cent.

The remaining two covariates, the growth proxy and months on treatment, are missing at rates unrelated to susceptibility and can be used: adjusting for either leaves the coefficient at -0.81 or -0.82 against an unadjusted -0.85 , an attenuation of at most five per cent (Table S25). The seeding gap is robust to every covariate the deposit permits testing against; the file records no referring site and no processing batch, so those cannot be tested at all. The association between resistance and a low starting inoculum is real, large and unexplained.

The two surviving associations sit in the 15-day panel, where the mechanism places them. By 60 days the cultures are 38-fold denser, the spread of starting densities has more than halved (interquartile range 1.00 to 0.42 log₁₀), and the fraction of isolates short of headroom falls from 15.2 to 3.3 per cent. The confound is a property of a thin assay, and it thins out when the assay is not. The two panels are columns on the same spreadsheet rows (210 isolates appear in both), so every comparison between them is within-isolate: 26 isolates lose their headroom shortfall between panels and none acquires one (exact McNemar $p = 3.0 \times 10^{-8}$ against an unpaired p of 2×10^{-5} (27)), eleven leave the floor and none joins it ($p = 9.8 \times 10^{-4}$), and every isolate whose headroom changes gains (Wilcoxon signed-rank $p = 1.3 \times 10^{-33}$ (28)). The interquartile range of starting density falls by 0.58 log₁₀.

It does not vanish, and the ordinal treatment shows where it goes. The one-odds-ratio-for-both-cuts assumption holds throughout the 15-day panel but fails at 60 days for starting density at the deepest endpoint (Brant $p = 0.0031$; likelihood-ratio test against a released fit $p = 0.0062$). Let that coefficient differ between the two cuts and the whole effect lands on the upper one: OR 0.493 for clearing medium (0.336 to 0.724, $p = 0.0003$) against 0.951 for clearing low (0.647 to 1.396, $p = 0.80$). A multinomial fit, assuming no ordering at all, agrees (high against low, relative risk ratio 0.59, $p = 0.028$; medium against low, 1.22, $p = 0.39$). In the denser panel the starting density no longer decides who is called low. It still decides who can be called high.

Extended results for article section 5: One written protocol does not give every laboratory the same depth to look down

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

The starting densities in Table S5 are the mean of quantified readings at day 0 or day 1 at the 100 μ L plating, pooled over that laboratory's arms. Institute A deposits no day-zero reading at all, and among treated flasks only C and D do, so for four of the six the figure rests partly on readings taken after twenty-four hours of drug, by which point the ten-times-MIC arms have already lost between 0.7 and 2.4 $\log_{10}$. A day-zero comparison across all six is therefore not available; re-running the whole analysis on a day-one exposure for every laboratory leaves the ordering, the area under the curve and the laboratory-level exact test unchanged.

Across all 30 laboratory-by-arm cells the Tobit estimate and the imputation estimate never differ by more than 0.006 $\log_{10}$ per day, which says the fit is not an artefact of the arithmetic used to reach it, not that the sub-floor assumption is safe: both estimators assume the same normal distribution below the floor, and nothing here can check that from the inside.

The duration endpoint behaves differently on the same flasks. What it records is not clearance and not sterilisation but a **first observed crossing below the assay floor**, and the distinction matters: most series that cross later read above the floor again, including all five untreated series that cross. The plating volume decides the count. At the 100 μ L quadruplicate plating, the most sensitive of the four, 29 of 72 treated flasks ever fall below the floor: institutes B and C record crossings in every arm, institute A in three of its four, institutes D, E and F in none (Fig. S1A). For half the laboratories the first-crossing time is right-censored throughout: the flask never crossed, so all that is known is that the crossing, if it comes, comes after the last visit.

That split is a property of the plating, not the laboratories. Counting a crossing on any of the four platings gives 48 of the same 72 treated flasks, and every laboratory records at least one: D six of its twelve, E one, F ten. The analysis keeps the single most sensitive plating so the endpoint means the same thing everywhere, and reports here what the other choice would have shown.

Starting density separates the flasks that ever crossed from those that never did with an area under the curve of 0.974, a description quoted without a p-value on purpose: the comparison looks like 67 flasks, but it is three laboratories against three, and the between-laboratory evidence in this deposit is six clusters throughout (Limitations). The exact laboratory-level test returns 0.10, the smallest value the design can return, and once the treatment arm is held fixed there is no within-laboratory comparison left to fall back on.

The Cox fits are reported for the same reason and with the same restraint. On laboratory alone, institutes D, E and F carry hazard ratios of 0.20, 0.21 and 0.20. Add starting density and those move to 0.34, 0.36 and 0.62, and concordance rises from 0.896 to 0.930 (Fig. S1B). Institute C moves the other way, from 3.0 to 5.3, and it is also the fastest killer, which is how a laboratory that genuinely kills faster ought to behave. No p-value is attached to any of these terms; the tested covariate is constant within cluster, and with six laboratories and six parameters the cluster-robust covariance is singular by construction, fitting it anyway returns a smaller p than the one it was meant to correct. A coefficient that moves on adjustment says how far starting density and laboratory identity overlap in this design; it is not evidence that one explains the other.

Where an interval is quoted from that fit it is a cluster bootstrap and not the model's own. For the starting-density coefficient the laboratory-clustered interval is -1.11 to -0.51 , half again as wide as the model's -0.91 to -0.48 and two and a half times the flask-clustered -0.81 to -0.56 ; quoting the model's interval is wrong in an unpredictable direction, not conservative. For the laboratory contrasts the flask-clustered interval is quoted instead, since the laboratory bootstrap keeps each laboratory's flasks together and the contrast barely moves. The crossings behind each laboratory's coefficient: 43 of 48 series in institute C against 1 of 48 in institute E.

Killing here is not sustained. Take the 64 treated series carrying at least three quantified readings at the most sensitive plating volume: in 48 the final step is not a decline, and 44 end more than one $\log_{10}$ above their own lowest reading, median rebound $2.17 \log_{10}$, largest 5.54 . Seventeen of the 32 flasks that fell below the floor at the most sensitive plating (29 treated, 3 untreated) were detectable again at a later visit, 14 of them in treated arms.

Across all four platings, 84 of the 140 series that cross return above the floor, 60 per cent, and the return is quick: a median of four days, with 45 per cent of those back within three. Those 360 series come from 90 flasks, four platings each, so they are not 360 independent observations; counted by flask, 53 of the 90 contain a crossing series (48 treated, 5 untreated), and 42 of those contain a returning one. A crossing below the assay floor in this deposit is usually a transient rather than an endpoint.

Killing itself is real and dose-dependent: at one times the inhibitory concentration five of six laboratories record net growth under moxifloxacin, between -0.047 and $-0.214 \log_{10}$ per day; at ten times all six record net decline.

Two consequences follow. First, the state below the floor is not absorbing: a series that drops below it does not stay there. Of 2,232 visit-to-visit transitions, 144 go from above to below and 95 go from below to above, so a series sitting below the floor leaves it at the next visit with probability 0.22, and that probability runs with drug pressure: 1.00 in untreated series, 0.37 to 0.92 at one times MIC, 0.07 to 0.18 at ten times, which says the event is at least partly a property of the plate. Of the 360 series, only 56, 15.6 per cent, show the shape a survival model assumes, one crossing that holds (Table S26); 220 never cross at all, 65 cross and return, 19 oscillate more than once.

Second, the crossing time is never observed: it lies between the last visit above the floor and the first visit below it, and the naive treatment pins it to the later end, which biases every duration late by construction. Fitted as interval-censored data (29) at the most sensitive plating, the tenth percentile of the first crossing falls from 2.95 to 1.82 days and the twenty-fifth from 11.24 to 9.64; across the treated arms at all four platings the twenty-fifth percentile falls from 5.77 to 3.67 days, a 36 per cent shortening. The non-parametric estimate cannot show this: fitted on the half-open interval the data actually give, the Turnbull curve agrees with the naive Kaplan-Meier at every visit, to floating-point precision, because the crossing intervals are the visit gaps themselves. What the pinning costs is everything between visits, which is where the parametric fit puts the shortening. A duration read off the naive curve is too long, by about a day at the depths that matter.

What the adjustment establishes: a continuous measure of where the cultures began accounts for the laboratory term at least as well as the laboratory label does, and one institute resists even that. What can be tested, since flasks are exchangeable across laboratories under the null, is how much of each quantity the laboratory owns: 87.0 per cent of the variance in starting density (permutation $p < 0.0002$), 36.8 per cent of the within-arm kill rate ($p = 0.0018$), and the duration endpoint outright, since three laboratories produce none at all.

Extended results for article section 6: The same boundaries hold in an experiment the framework never saw

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

Every section so far tests the framework on the deposits it was built from, all *Mycobacterium tuberculosis*. Of the 45 deposits whose series could be read in full, 18 permit reachability and identifiability to be computed once the floor is established on the tiers this paper uses for its own: stated by the depositor, derived from a recorded plated volume, or inferred from a pile-up of counts on a plate-plausible value. Across those, the condition the boundaries imply, that no series may present as a measurement a reduction deeper than its own floor allows, holds in 2,210 of 2,210 series. Ten of the eighteen name their organism, six distinct species; the other eight do not, and are described in the data release rather than here. One is the held-out deposit below, fetched independently under a second identifier, so this corpus-wide check and the cold test are not two confirmations.

Only one deposit was opened after every boundary and threshold was fixed. That it is *E. coli* in a hollow-fibre system rather than *M. tuberculosis* in a flask makes it a harder test, not a softer one: a rule fitted to tuberculosis time-kill data could be expected to fit more of it; what it has no right to survive is a different organism, drug class and apparatus. The boundaries are derived from definitions, so they predict they will.

Extended results for article section 8: One culture, two plated volumes, two tolerance labels

Material held back from the article for length. It is reported here rather than dropped, because each paragraph answers a question a reader of that section may reasonably ask.

It is *Escherichia coli* and not *M. tuberculosis* by design, not convenience. Reachability and identifiability are arithmetic on a plated volume, so they do not know what organism is on the plate; what an organism decides is whether the experiment can be run at all. Three seeding densities crossed with two plated volumes and six sampling times is 216 plates that must be counted from one exposure, and in *M. tuberculosis* that means three to six weeks to colonies, inside containment, per replicate. Ciprofloxacin drives *E. coli* to the floor within a working day, which makes the floor reachable in every arm and the prediction falsifiable in a week rather than a year. ATCC 25922 is the quality-control reference strain the standard itself names, so the demonstration runs in the one organism whose assay geometry the standard defines.

The gain the pipette buys should be exactly one log, because the floor at the pooled 100 μ L plating is one tenth of the floor at the pooled 10 μ L plating. The nine paired gains run 0.87 to 1.19 (Table S9). The difference is not the drug and not the flask: each plating measures its own starting density from the same culture, and the two measurements disagree, 425,000 against 565,000 per mL in one flask, 255,000 against 375,000 in another. That disagreement is this paper's own argument turned on its own experiment: No is a measurement with error, and the headroom computed from it inherits that error. Reported per flask, the gain never falls below 0.87 log10, so the four-log endpoint moves from unreportable to reportable in every one of the three cultures at the standard inoculum.

Read the experiment the way a laboratory would actually run it, with each plating series carrying the starting density it measured for itself, and the count is nine of 54 rather than six. The three extra are not the pipette: they are the two platings disagreeing about how much culture went in, differing by 0.74- to 1.53-fold across the nine flasks, median 1.12. Six is what the plated volume does on its own; nine is what a laboratory would see, because in a real experiment the starting density is measured too, and it is measured with error.

Supplementary figures

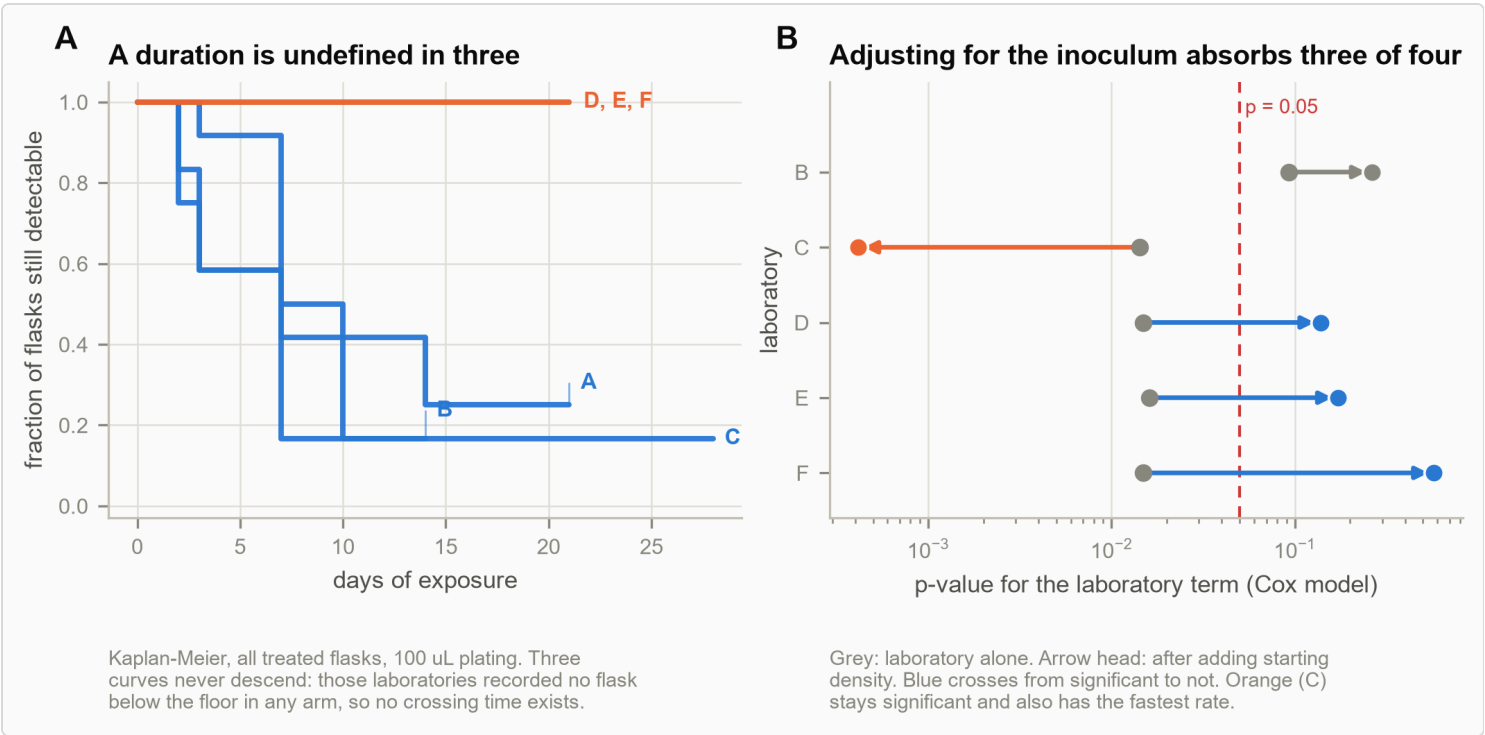


Figure S1. The descriptive survival picture behind Figure 2. The same six laboratories, one written protocol, one stock of *M. tuberculosis* H37Rv, moxifloxacin at ten times the minimum inhibitory concentration. (A) Kaplan–Meier curves for time to the first observed crossing below the assay floor at the 100 μ L plating. Three curves never descend: those laboratories recorded no flask below the floor in any arm at that plating, so their first-crossing times are right-censored throughout. (B) The model-based p-value attached to each

laboratory in a descriptive Cox model before (grey) and after (arrow head) starting density is added, showing how far the two predictors overlap. Institute A is the reference, so the model carries five laboratory terms; four of them move toward $p = 1$ on adjustment (B from 0.093 to 0.264, D from 0.015 to 0.138, E from 0.016 to 0.172 and F from 0.015 to 0.576), while institute C moves the other way, from 0.014 to 0.00041. Those p-values are plotted because they are what moves, not because they are valid: the laboratory label is constant within its own cluster and there are six clusters, so no between-laboratory test is available and none is claimed. Blue marks a term crossing the conventional threshold, which is a description of the movement and not a finding. Both panels appear here, rather than in Figure 2, because each requires its own disclaiming legend. Colour, and the split between laboratories that ever crossed and those that never did, are shared with Figure 2 by construction: both figures read the assignment from one function.

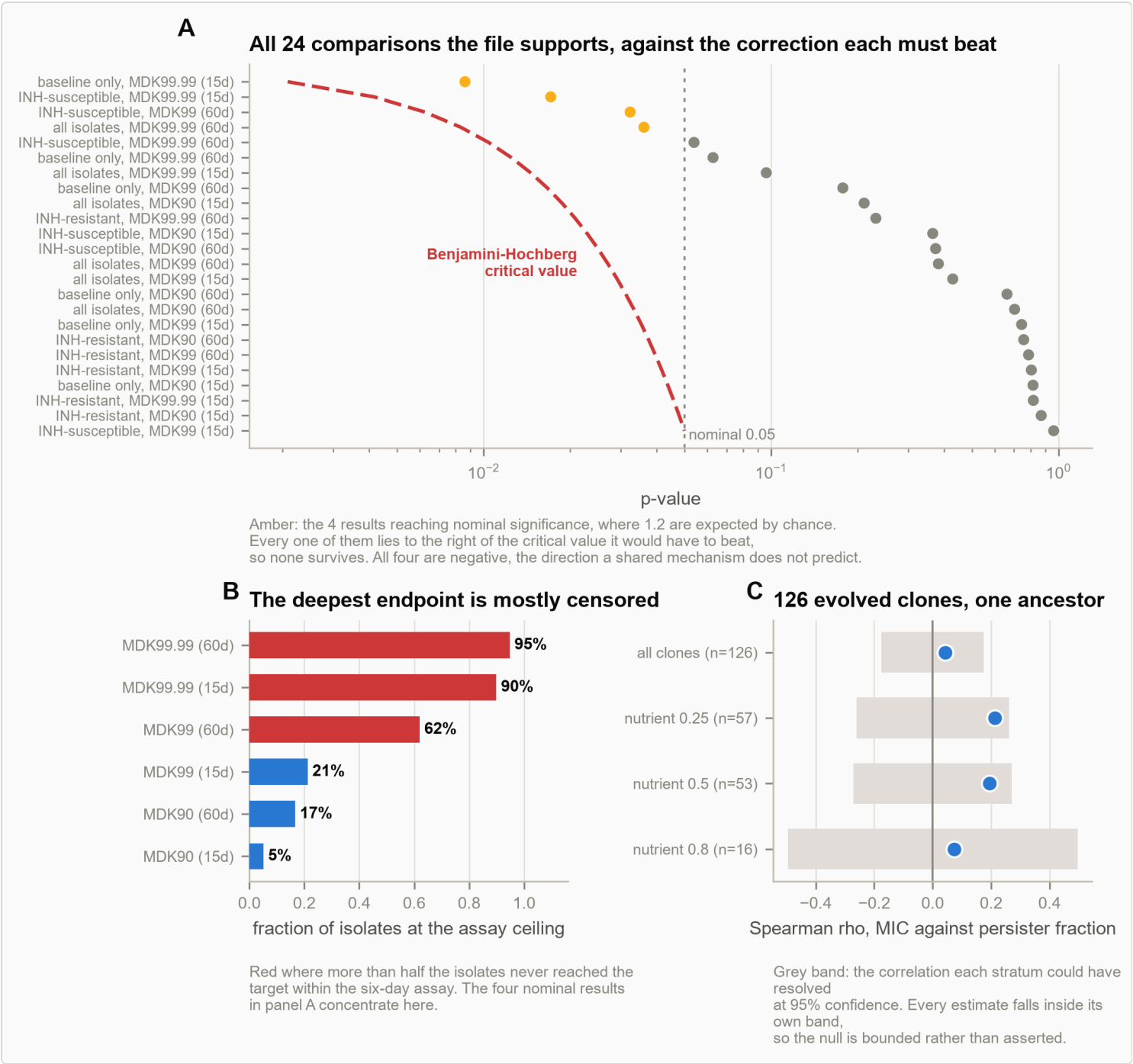


Figure S2. Resistance and tolerance occupy separate axes, in two designs. (A) All 24 comparisons between the concentration axis and the duration axis that the 217-isolate file supports, each against the Benjamini–Hochberg critical value it would have to beat. Amber marks the four reaching nominal significance; none survives. **(B)** The censoring behind them: the fraction of isolates at the assay ceiling rises with endpoint depth, and the nominal hits concentrate where censoring is heaviest. **(C)** The same question in 126 evolved *E. coli* clones, by nutrient stratum.

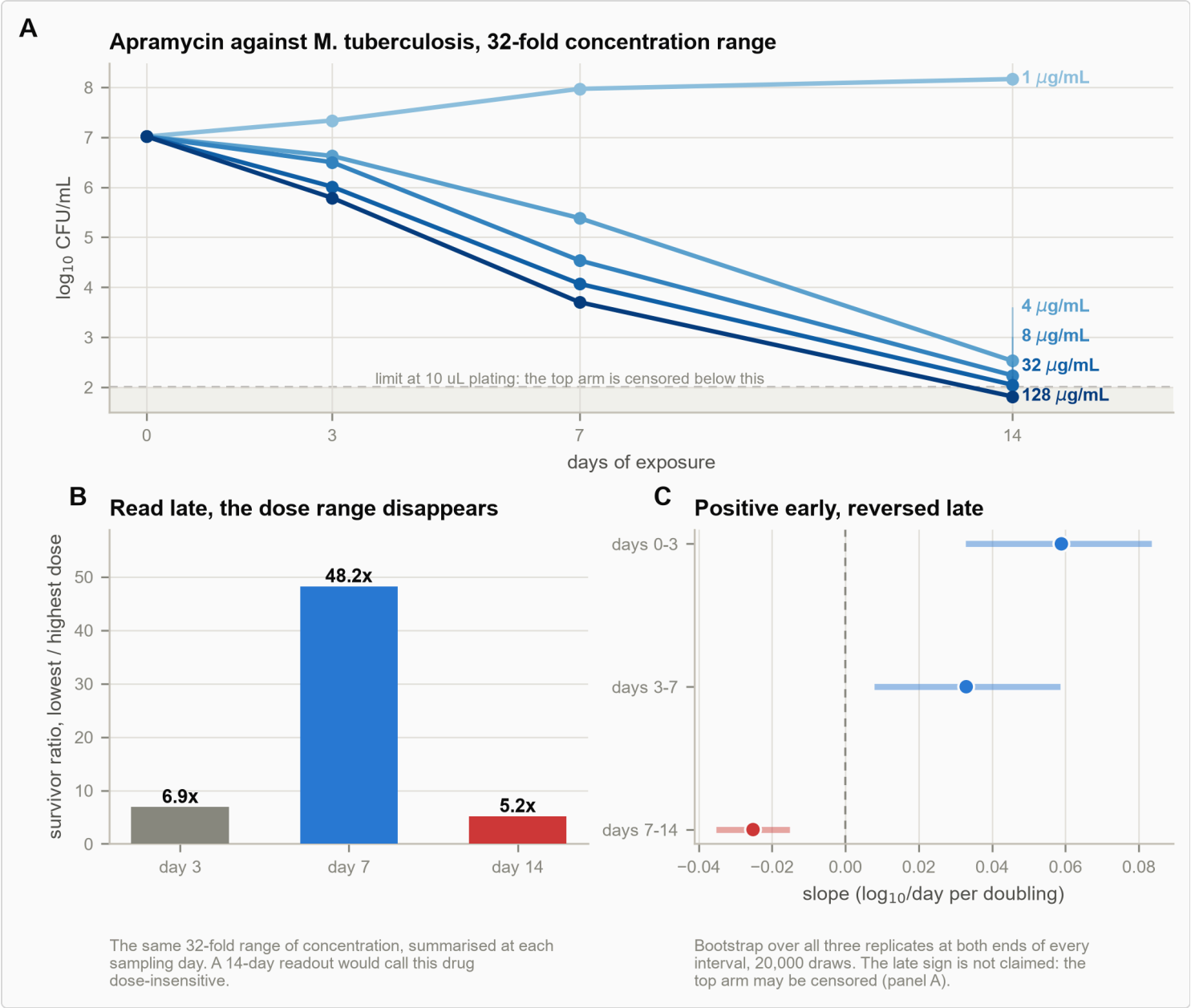


Figure S3. A late endpoint cannot resolve a 32-fold concentration range. Apramycin against *M. tuberculosis*, five concentrations, triplicate counts. **(A)** The trajectories, with an assumed floor of 100 CFU/mL (the value 10 μ L plating would give), drawn as a band; this deposit records no plated volume, so the band is an assumption and is drawn to show what the assumption costs; by day 14 the highest arm lies within it, so its apparent rate over the final interval is a lower bound. **(B)** The ratio of survivors between the lowest and highest

of the four compared concentrations, 4 and 128 µg/mL, at each sampling day, rising to 48.2-fold at day 7 and collapsing to 5.2-fold at day 14. (C) The concentration slope fitted separately in each interval from a replicate-level bootstrap, 20 000 draws.

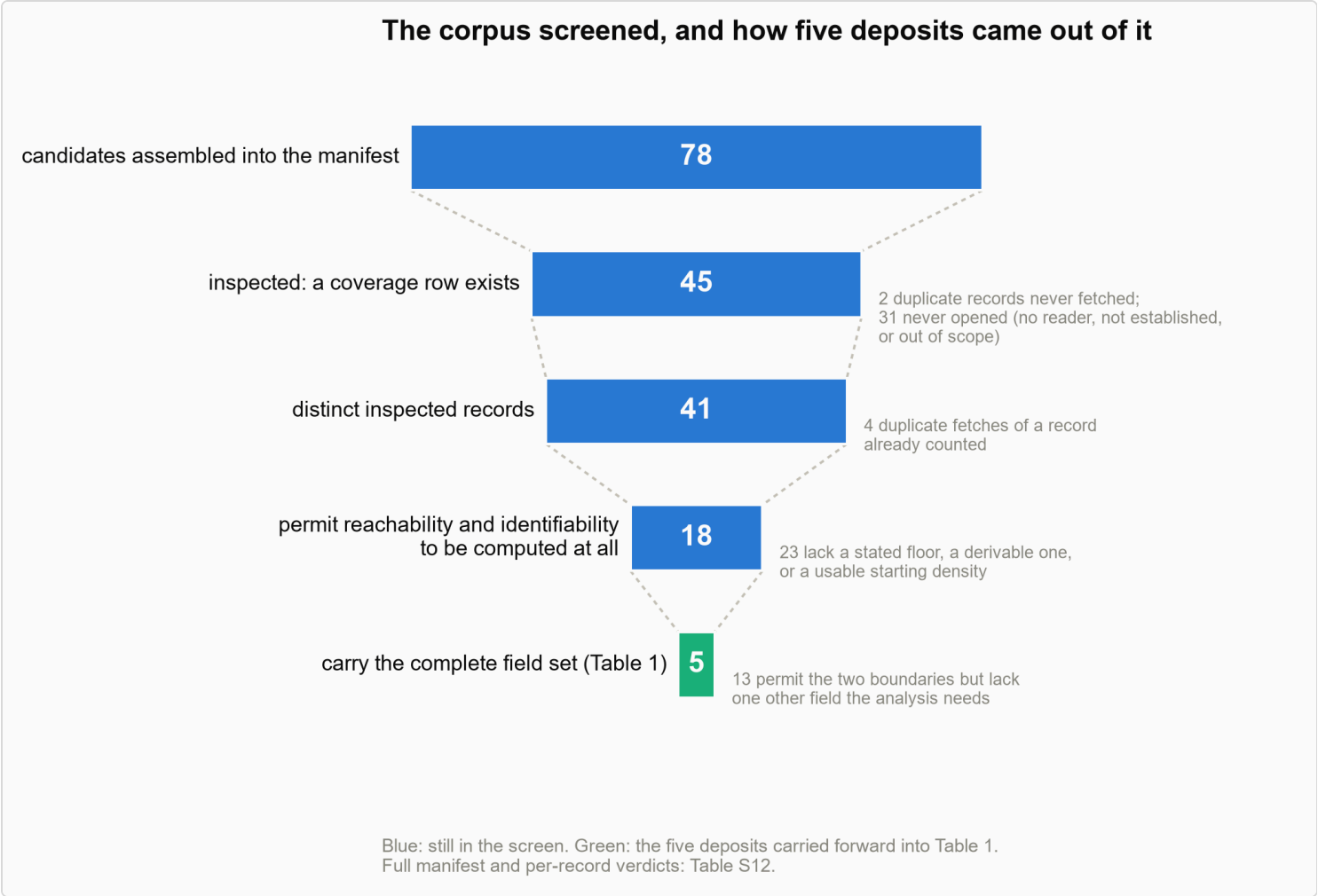


Figure S4. The corpus screened, and how five deposits came out of it. A funnel of the literature screen described in Methods, from the 78 candidate time-kill datasets assembled to the five carried forward into Table 3. Each stage names what left the funnel to produce the next count. Blue: still in the screen. Green: the five deposits reanalysed in the article. Full manifest and per-record verdicts: Table S1.

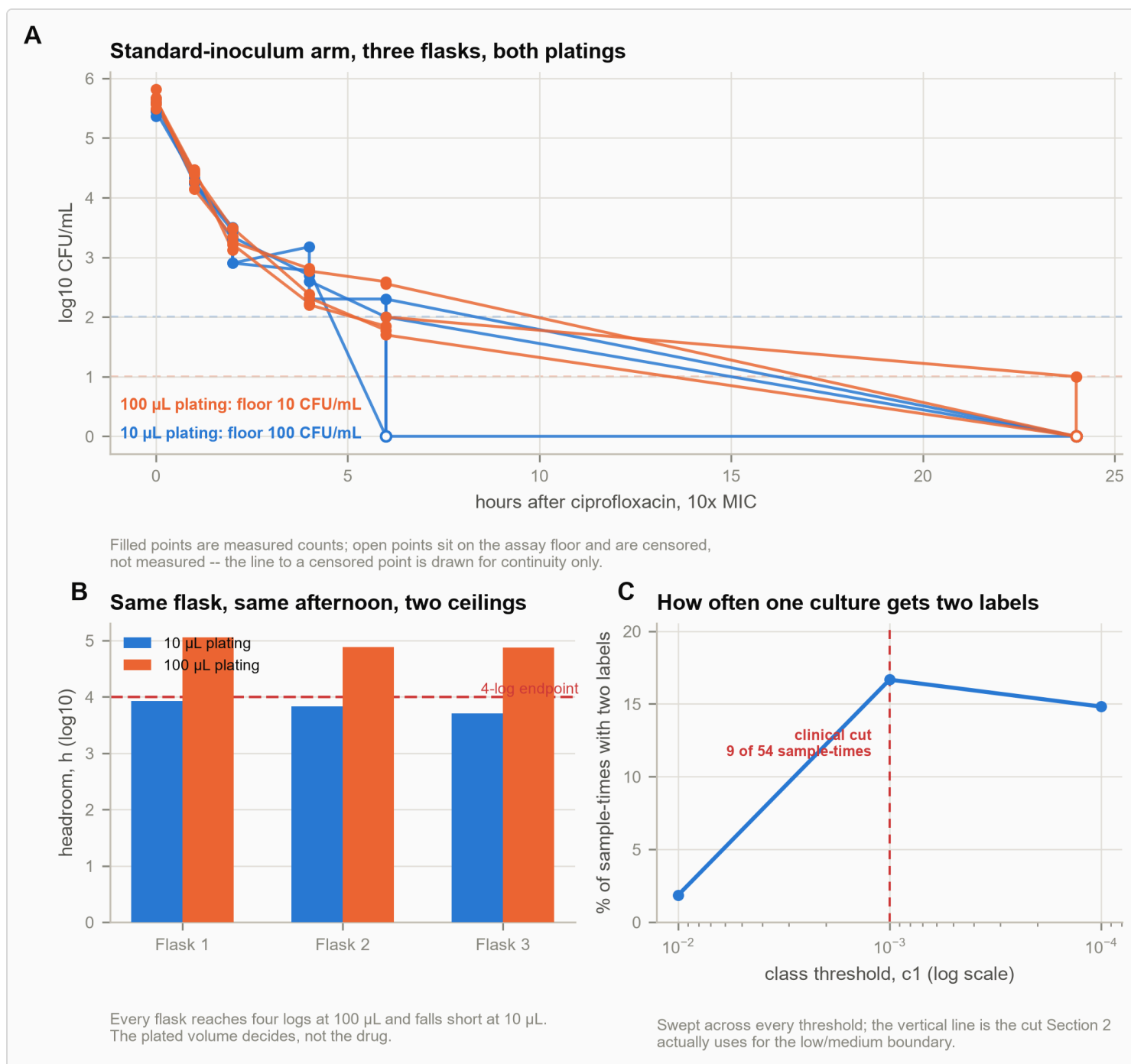


Figure S5. One culture, two plated volumes, two tolerance labels. The prospective experiment's standard-inoculum arm, three flasks, plated at 10 and 100 µL from the same dilution series. **(A)** Kill curves at both platings; filled points are measured counts, open points sit on the assay floor and are censored. **(B)** Headroom per flask at each plating against the four-log endpoint: every flask reaches it at 100 µL and falls short at 10 µL. **(C)** The threshold sweep behind Table S10: the share of sample-times at which the two platings' recorded fractions fall on opposite sides of a class cut, as the cut moves; the marked line is the cut the clinical classification of Section 2 uses.

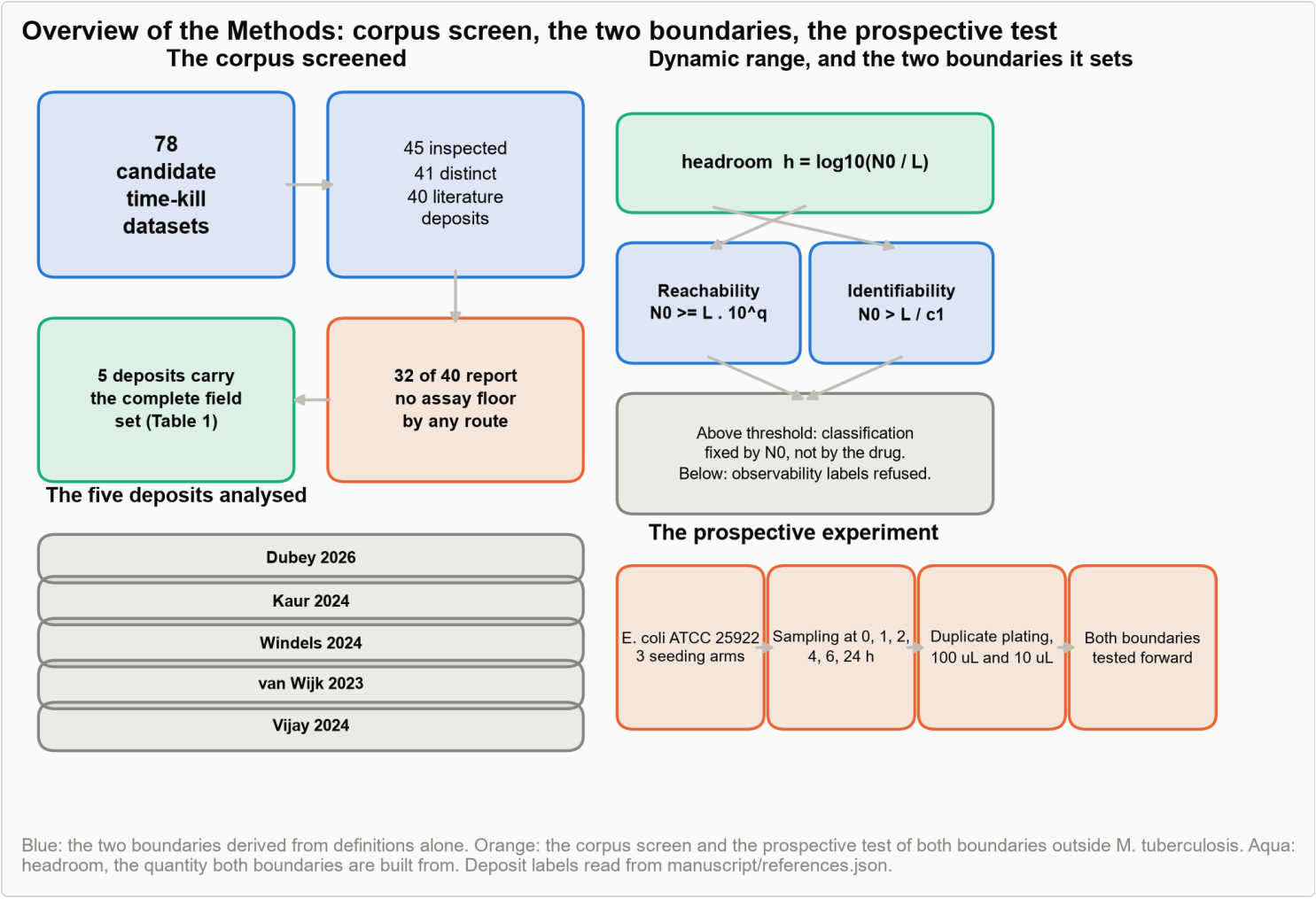


Figure S6. Overview of the Methods. The corpus screen that produced the five analysed deposits, the two boundaries derived from the assay's own definitions, and the prospective experiment that tests both forward. Deposit labels and every count are read from the same tables and reference list the Methods cite (Table 3, Table S1).

Supplementary tables

Table S1. The screen, stage by stage.

Stage	n
candidates assembled into the manifest	78
marked a duplicate record, never fetched	-2
never opened (no reader, not established, or out of scope)	-31
inspected: a coverage row exists	45
of those, a duplicate fetch of a record already counted	-4
distinct inspected records	41
of those, this paper's own prospective experiment	-1
DISTINCT LITERATURE DEPOSITS INSPECTED	40
of those, stating no assay floor by any route	32

Note. Counts in the Abstract, the Introduction, the Methods and the Discussion are read off this table. A negative row is a deduction from the line above it. "Distinct literature deposits inspected" is the denominator for the proportion stating no assay floor: it excludes duplicate fetches of records already counted, and it excludes this paper's own prospective experiment, which is not a screened literature deposit. The 33 candidates that were never opened are not characterised here or anywhere, and the proportion is not extrapolated to them.

Table S2. The deposited tolerance class at day 5 after 15 days of prior culture is a threshold on the recorded surviving fraction.

Recorded class	n	Min fraction	Max fraction	Predicted from cuts	Disagreements
Low	33	3.8×10^{-6}	4.7×10^{-4}	Low	0
Medium	124	1.0×10^{-3}	1.0×10^{-2}	Medium	0
High	46	2.7×10^{-2}	1.0×10^1	High	0
All usable	203	-	-	-	0

Note. With no overlap between classes: low below 10^{-3} , medium from 10^{-3} to 10^{-2} inclusive, high above 10^{-2} . Applying those cuts reproduces every usable class in the file. This matters because the argument that follows is about the fraction the assay recorded, not about an independent clinical judgement.

Table S3. Every analysis set in this paper, and the exclusion that produced it.

Deposit	Panel	Stratum	Stage	n	Excluded here
Vijay clinical	15-day	all isolates	rows in the deposit	217	-
Vijay clinical	15-day	all isolates	tolerance label present	217	-
Vijay clinical	15-day	all isolates	label is Low, Medium or High (drops 'MDR')	203	-14
Vijay clinical	15-day	all isolates	starting and day-5 readings present	203	-
Vijay clinical	15-day	all isolates	susceptibility is IS or IR	203	-
Vijay clinical	15-day	all isolates	growth proxy present (association family)	202	-1
Vijay clinical	15-day	baseline only (0M)	rows in the deposit	174	-
Vijay clinical	15-day	baseline only (0M)	tolerance label present	174	-
Vijay clinical	15-day	baseline only (0M)	label is Low, Medium or High (drops 'MDR')	168	-6
Vijay clinical	15-day	baseline only (0M)	starting and day-5 readings present	168	-
Vijay clinical	15-day	baseline only (0M)	susceptibility is IS or IR	168	-
Vijay clinical	15-day	baseline only (0M)	growth proxy present (association family)	167	-1
Vijay clinical	60-day	all isolates	rows in the deposit	217	-
Vijay clinical	60-day	all isolates	tolerance label present	211	-6
Vijay clinical	60-day	all isolates	label is Low, Medium or High (drops 'MDR')	197	-14
Vijay clinical	60-day	all isolates	starting and day-5 readings present	197	-
Vijay clinical	60-day	all isolates	susceptibility is IS or IR	197	-
Vijay clinical	60-day	all isolates	growth proxy present (association family)	196	-1

Deposit	Panel	Stratum	Stage	n	Excluded here
Vijay clinical	60-day	baseline only (0M)	rows in the deposit	174	-
Vijay clinical	60-day	baseline only (0M)	tolerance label present	168	-6
Vijay clinical	60-day	baseline only (0M)	label is Low, Medium or High (drops 'MDR')	162	-6
Vijay clinical	60-day	baseline only (0M)	starting and day-5 readings present	162	-
Vijay clinical	60-day	baseline only (0M)	susceptibility is IS or IR	162	-
Vijay clinical	60-day	baseline only (0M)	growth proxy present (association family)	161	-1
Vijay clinical	15-day	other sets quoted	isolates with a starting density	217	-
Vijay clinical	60-day	other sets quoted	isolates with a starting density	210	-7
Vijay clinical	15-day	concentration-duration family	all isolates with an inhibitory concentration	193	-
Vijay clinical	60-day	concentration-duration family	all isolates with an inhibitory concentration	187	-
Vijay clinical	15-day	concentration-duration family	INH-susceptible with one	119	-
Vijay clinical	60-day	concentration-duration family	INH-susceptible with one	117	-
Vijay clinical	15-day	concentration-duration family	baseline only with one	162	-
Vijay clinical	60-day	concentration-duration family	baseline only with one	156	-
ERA4TB six-laboratory	-	-	readings in the file	2775	-
ERA4TB six-laboratory	-	-	excluding inoculum controls	2751	-24
ERA4TB six-laboratory	-	-	flasks (institute x arm x replicate)	90	-
ERA4TB six-laboratory	-	-	treated flasks	72	-18

Deposit	Panel	Stratum	Stage	n	Excluded here
ERA4TB six-laboratory	-	-	treated flasks with a usable starting density	67	-5
ERA4TB six-laboratory	-	-	laboratory-by-arm cells	30	-
ERA4TB six-laboratory	-	-	cells with a fitted kill rate	29	-1
ERA4TB six-laboratory	-	-	series under the plating key (flasks x 4 platings)	360	-
ERA4TB six-laboratory	-	-	treated series	288	-72
ERA4TB six-laboratory	-	-	cross-laboratory pairs in the same arm	191	-
ERA4TB six-laboratory	-	derived units	treated series with a quantified day-0 or day-1 reading in their own plating	262	-26
ERA4TB six-laboratory	-	derived units	of those, not already below their own floor at day zero (the descriptive Cox set)	261	-1
ERA4TB six-laboratory	-	derived units	cross-laboratory pairs, strict rate separation	100	-
ERA4TB six-laboratory	-	derived units	flasks contributing those pairs	42	-
ERA4TB six-laboratory	-	derived units	laboratory-by-arm-by-volume cells with a measured start	56	-
ERA4TB six-laboratory	-	derived units	treated series with three or more quantified readings at 100 uL	64	-
ERA4TB six-laboratory	-	derived units	flask units in the variance decomposition (all arms plus inoculum)	85	-
ERA4TB six-laboratory	-	derived units	untreated day-zero readings at 100 uL (4 laboratories x 3 flasks)	12	-
ERA4TB six-laboratory	-	derived units	below-limit flags in the analysis set	498	-
ERA4TB six-laboratory	-	derived units	readings in the kill-rate analysis set	2580	-171
Dubey hollow fibre (held out)	-	-	cultures with a measured day-zero density	20	-

Note. The manuscript quotes a dozen different denominators, each correct for its own analysis; this is where a reader checks which is which. The MDR tolerance label is a fourth, unordered category and is dropped wherever

an ordered outcome is fitted; the growth proxy is missing for one isolate; and five treated flasks in the six-laboratory deposit carry no usable starting density, which is why a comparison over 72 flasks is reported on 67. Whether these exclusions are plausibly ignorable is tested in Table S17. The last rows of each block cover sets counted in units other than isolates or flasks -- pairs, flags, cells, readings -- because a reader meeting one of those figures in the text needs somewhere to look it up too.

Table S4. The family of 8 tests between the deposited tolerance label and its candidate determinants (BH: Benjamini-Hochberg; OD: optical density).

Predictor	Prior culture	Reading day	For starting density	n	Odds ratio (95% CI)	p	Survives BH	Prop. odc
time to OD 0.4	15 d	day 5	adjusted	202	1.096 (1.03-1.16)	0.0030	yes	0.
resistance	15 d	day 5	unadjusted	202	2.317 (1.30-4.12)	0.0042	yes	0.
time to OD 0.4	60 d	day 2	adjusted	196	0.933 (0.87-1.00)	0.0470	no	0.
time to OD 0.4	15 d	day 2	adjusted	202	1.068 (1.00-1.14)	0.0618	no	0.
resistance	15 d	day 2	unadjusted	202	1.295 (0.66-2.54)	0.4529	no	0.
time to OD 0.4	60 d	day 5	adjusted	196	0.976 (0.91-1.04)	0.4605	no	0.
resistance	60 d	day 2	unadjusted	196	0.836 (0.47-1.48)	0.5400	no	0.
resistance	60 d	day 5	unadjusted	196	1.111 (0.63-1.95)	0.7150	no	0.

Note. Fitted as proportional-odds ordinal logistic regression and corrected together by the Benjamini-Hochberg procedure at a false discovery rate of 5%; the "survives BH" column is that correction. 2 survive. An odds ratio above one means higher odds of a higher tolerance class; time to an optical density (OD) of 0.4 is in days and runs inversely to growth rate, so above one there means *slower* growth accompanies a higher class. The proportional-odds column is a Brant test per predictor; the assumption holds for every predictor in this family. Starting density enters the adjusted fits as a covariate rather than as a member of the family, and it is where proportional odds fails, at 60 days at the deepest endpoint; that failure and the released fit are reported in Section 4. The final column repeats each test on the baseline isolates, one per patient by construction, which is where the resistance association stops clearing its corrected threshold. The deposit carries no patient identifier, so no standard error here can be clustered on the true grouping and every interval in this table is model-based; the baseline-isolate column is the sensitivity analysis that stands in for clustering, and Table S6 records what each conclusion is worth once it is applied.

Table S5. Moxifloxacin at ten times MIC.

Lab	Starting density (log10 CFU/mL)	Reading day	Kill rate (log10/day)	95% profile interval	By imputation	Readings
A	2.95	day 1	0.119	0.084 to 0.160	0.125	
B	3.16	day 0, day 1	0.139	0.097 to 0.191	0.140	
C	4.02	day 0, day 1	0.464	0.408 to 0.528	0.464	
D	4.69	day 0, day 1	0.104	0.077 to 0.132	0.104	
E	4.95	day 0, day 1	0.090	0.073 to 0.107	0.090	
F	6.53	day 0, day 1	0.223	0.191 to 0.254	0.223	

Note. Every laboratory yields a rate; three record no crossing below the assay floor in any arm at the 100 uL plating. The final column counts first observed crossings, which are not clearances: across the deposit 60% of the series that cross read above the floor again at a later visit. The two censoring estimators agree to 0.003 log10 per day. The starting density is the mean of quantified readings at day 0 or day 1 at the 100 uL plating, pooled over all of that laboratory's arms; the reading-day column says which visits contributed. Institute A deposits no day-zero reading at all, and among treated flasks only institutes C and D do, so for the rest this figure rests partly on readings taken after 24 hours of drug. The turbidity-standard analysis and Table S18 instead use untreated day-zero readings only, so the two tables are not expected to match.

Table S6. Every conclusion this paper draws from the two primary deposits, against uncertainty recomputed at the level the observations are actually independent.

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead	
4	the difference between panels in the growth association is itself supported (interaction $p = 0.0072$)	unknown isolate-panel (each of 210 isolates counted once per panel)	210 isolates	not recomputable	:
5	starting density separates flasks that ever crossed below the assay floor from those that never did (AUC 0.974, $p = 1.2e-10$)	67 flask	6	exact laboratory-level test (3 crossed vs 3 did not); cluster bootstrap over laboratories; permutation of crossing within laboratory, and within laboratory AND treatment arm	:
5	in a Cox model, institutes D, E and F carry $p = 0.015, 0.016, 0.015$, which adjustment for starting density moves to 0.138, 0.172, 0.576	67 flask	6	cluster-robust sandwich on six clusters	:
5	the log-rank test separates the six laboratories on time to first crossing	67 flask	6	none available: the grouping variable is the cluster	:
5	institute C remains distinguishable after adjustment (HR 5.27, $p < 0.001$)	67 flask	1	none available	:
1	all 33 short isolates are at the ceiling; 88.0% of the rest are (Fisher $p = 0.030$)	217 isolate	174 patients min	none: the null is arithmetically impossible	W
5	at ten times MIC of moxifloxacin the kill rate spans 5.2-fold between laboratories (0.090 to 0.464)	6 laboratory-arm cell	6	cluster bootstrap over whole laboratories	
5	at one times MIC five of six laboratories record net growth	6 laboratory-arm cell	6	Jeffreys interval on six clusters	
5 / 7	the laboratory effect lands on the duration endpoint and not on the rate; the rate is 'far more reproducible'	42 flask	6	permutation of the laboratory label across flasks, within arm for the rate	
6	the deepest demonstrable kill differs by 2.33 log10 between laboratories at one plating volume	12 day-zero reading	4	cluster bootstrap over whole laboratories	

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead	
7	34.6% of 191 cross-laboratory pairs are inversions (95% CI 28.1-41.5)	191 pair of flasks	6	cluster bootstrap over whole laboratories; also over flasks within laboratory; also delete-one-laboratory jackknife	
7	under the strictest rate separation the inversion rate is 23.0% of 100 pairs (95% CI 15.6-31.9)	100 pair of flasks	6	cluster bootstrap over whole laboratories	
1	33 of 217 isolates (15.2%) lack the headroom for a 4-log endpoint	217 isolate	174 patients min	baseline-only recount	:
1 / 2	18 isolates rest on the floor at 15 days and 6 at 60	217 + 210 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	exact McNemar on the paired binary	:
10	no MIC-MDK association survives Benjamini-Hochberg (supplementary text, not a numbered section of the article)	6 tests isolate	174 patients min	baseline-only family of six	:
2	18 isolates at the floor; their survival span IS their inoculum span	217 isolate	174 patients min	baseline-only recount	:
2	185 determinable, 12 forced by the inoculum, 6 undecidable	203 isolate	174 patients min	baseline-only reclassification	:
2	at 60 days the observability counts are 191, 6 and none, 'because the cultures are denser and few readings reach the floor'	203 + 197 isolate-panel (each of 210 isolates counted once per panel)	197 isolates	exact McNemar on the paired binary	:
2 / 4	the deep endpoint is unreachable for 26.2% of resistant against 7.6% of susceptible isolates	217 isolate	174 patients min	baseline-only Fisher exact	:
4	resistant isolates enter the assay ten-fold lower	217 isolate	174 patients min	baseline-only Mann-Whitney	:
4	the resistance-tolerance association attenuates once starting density enters the model, and its interval then spans zero	202 isolate	174 patients min	baseline-only refit	:

Section	Conclusion as originally stated	Units treated as independent	Independent clusters	Method used instead	
4	growth state predicts the tolerance class and survives adjustment (beta = +0.027, p = 0.0025)	202 isolate	174 patients min	baseline-only refit	:
4	resistant isolates do not grow measurably more slowly (p = 0.24)	217 isolate	174 patients min	baseline-only Mann-Whitney	:
4	the confound thins between panels: 15.2% short of headroom at 15 days against 3.3% at 60	217 + 210 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	exact McNemar on the paired binary	:
4	the 60-day cultures are denser, so headroom is larger there	420 treated as independent isolate-panel (each of 210 isolates counted once per panel)	210 isolates	Wilcoxon signed-rank on the within-isolate change	:
4	the spread of starting densities more than halves between panels (IQR 1.00 to 0.42 log10)	420 isolate-panel (each of 210 isolates counted once per panel)	210 isolates	paired bootstrap resampling whole isolates	:
5	of 64 treated series the final step is not a decline in 48 (75%)	64 series (one per treated flask)	6	cluster bootstrap over whole laboratories	:
5	44 of 64 series end more than one log10 above their own nadir	64 series (one per treated flask)	6	cluster bootstrap over whole laboratories	:
5 / Limitations	87.0% of the variance in flask starting density lies between laboratories	85 flask	6	permutation of the laboratory label across flasks	:
6	including the choice of plated volume widens the spread in measurable depth to 4.40 log10	up to 4 volumes x 5 laboratories laboratory-by-volume channel	6	no resampling required for the volume term	:
7	the criterion $D_A/D_B > b_A/b_B$ calls 83.8% of pairs correctly	191 pair of flasks	6	cluster bootstrap over whole laboratories	:
Methods	83 of 498 below-limit flags (16.7%) are contradicted by another plating of the same sample at the same visit	498 flag (one reading)	6	cluster bootstrap over whole laboratories	:

Note. 20 survive unchanged, 6 survive with materially wider uncertainty, 5 do not survive, and 1 is withdrawn because its null is false before any data are seen. Each verdict applies to the claim as it was originally stated.

Three of the five failures are gone from the text entirely; for the other two a weaker statement is retained and is marked as such where it appears -- the Cox coefficient trade is now reported as hazard ratios with no p-value, and institute C as the laboratory whose crossings its starting density does not account for rather than as a tested contrast. The five that fail are all between-laboratory p-values computed on flasks: every flask in a laboratory shares a starting culture, so a comparison that looks like 67 flasks is six laboratories, and for a three-against-three split of six clusters the smallest attainable two-sided p is 0.10. They are deleted rather than corrected, because there is nothing to correct them to.

Table S7. Dubey et al. 2026, analysed cold.

Endpoint	N_reach (per mL)	Cultures	Unreachable	Per cent
1 log (90%)	100	20	0	0%
2 log (99%)	1,000	20	0	0%
3 log (99.9%)	10,000	20	0	0%
4 log (99.99%)	100,000	20	0	0%
5 log (99.999%)	1,000,000	20	5	25%
6 log (99.9999%)	10,000,000	20	20	100%

Note. The floor is derived from a stated 100 uL plated volume and corroborated inside the file: all 229 genuine counts are multiples of ten and the smallest is exactly ten. Starting densities are the 20 measured day-zero counts, not the nominal inoculum the Methods state.

Table S8. Every *Mycobacterium tuberculosis* deposit in the screened corpus whose assay floor could be established.

Drug or regimen	Floor is	<i>L</i> (CFU/mL)	Series	Median <i>h</i>	Short of 4 logs	Short of 5 logs	Deposit
pretomanid, Q203, and the combination	stated	20	21	5.60	0	0	BioStudies, 2026
24 regimen arms, sputum from pulmonary TB	derived	24	112	4.23	43	83	figshare 50649807
rifampin, isoniazid, streptomycin and others	derived	13	27	6.64	7	7	Europe PubMed Central, 2021
isoniazid, bedaquiline, Q203, pretomanid	inferred	200	123	2.85	123	123	figshare 39444559
ethionamide, isoniazid, alpipectir	inferred	100	27	4.10	11	27	Nat Commun, 2026

Note. And how deep its own series could see. Across the five, 184 of 310 series could not have demonstrated a four-log reduction however completely the drug worked, and 240 of 310 could not have demonstrated five. Headroom is $h = \log_{10}(N_0/L)$ and the floor is stated by the source, derived from a recorded plated volume, or inferred from a pile-up on the lowest reported value, as the column says. None of these deposits carries the analysis in the article; they are what the screen found in the same organism. The full accession for each deposit is in the corpus manifest released with the analysis code.

Table S9. The deepest reduction each culture in the prospective experiment could report, at each of its two platings.

Arm	Flask	N_0 at 10 μ L (per mL)	N_0 at 100 μ L (per mL)	h at 10 μ L	h at 100 μ L	Gain
High	1	3,600,000	4,050,000	4.86	5.91	+1.05
High	2	4,450,000	3,300,000	4.95	5.82	+0.87
High	3	4,450,000	3,950,000	4.95	5.90	+0.95
Low	1	58,000	78,000	3.06	4.19	+1.13
Low	2	42,500	65,000	2.93	4.11	+1.18
Low	3	45,500	50,000	2.96	4.00	+1.04
Mid	1	425,000	565,000	3.93	5.05	+1.12
Mid	2	340,000	380,000	3.83	4.88	+1.05
Mid	3	255,000	375,000	3.71	4.88	+1.17

Note. $h = \log_{10}(N_0/L)$ with L one colony in the pooled volume plated, taken at the lowest dilution the series was read at, which is where the floor is lowest and the reportable depth greatest. The two platings of one flask share a row: at the standard inoculum the 10 μ L plating ceiling runs 3.71 to 3.93 logs and the 100 μ L ceiling 4.88 to 5.05, so a four-log endpoint is unreportable at one plating and reportable at the other in the same culture. The gain column is close to the $\log_{10}(10) = 1.00$ that plating ten times the volume buys; it is not exactly 1.00 because each plating measures its own N_0 and the two measurements differ.

Table S10. How often one culture receives two different tolerance labels from its two platings, swept across the class threshold.

Class threshold c_1	Straddling, one N_0 per flask	Share	Straddling, each plating's own N_0	Share
10^{-2}	1 of 54	2%	1 of 54	2%
10^{-3}	6 of 54	11%	9 of 54	17%
10^{-4}	7 of 54	13%	8 of 54	15%

Note. That two platings report different fractions is arithmetic; that those fractions land either side of a cut is not, and this is the quantity that could have come out zero. At a cut of one per cent the window is nearly shut. At one in a thousand, where the clinical classification reanalysed here cuts its lowest class, it is one sample-time in nine. The two pairs of columns differ in one thing. The first holds the starting density to a single value per flask, so the only thing separating the two readings is the plated volume, which is what the argument claims. The second lets each plating carry the starting density it measured for itself, which is what a laboratory running both platings would have; the count rises because those two estimates of one culture disagree by 0.74- to 1.53-fold. The first column is the claim, the second is the practice, and neither is the other.

Table S11. Tolerance classification as posterior inference.

Panel	Isolate	NO (MPN/mL)	Recorded class	P(Low)	P(Medium)	P(High)	Verdict at 95%
15-day	8	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	43	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	49	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	50	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	61	610000.0	Low	1.0	0.0	0.0	confirmed
15-day	66	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	76	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	103	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	104	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	118	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	128	6100000.0	Low	1.0	0.0	0.0	confirmed
15-day	135	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	137	2300000.0	Low	1.0	0.0	0.0	confirmed
15-day	138	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	148	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	161	230000.0	Low	0.999	0.001	0.0	confirmed
15-day	167	23000.0	Medium	0.5331	0.4655	0.001	refused
15-day	186	6100000.0	Low	1.0	0.0	0.0	confirmed
60-day	43	610000.0	Low	1.0	0.0	0.0	confirmed
60-day	49	230000.0	Low	0.999	0.001	0.0	confirmed
60-day	50	230000.0	Low	0.999	0.001	0.0	confirmed
60-day	66	6100000.0	Low	1.0	0.0	0.0	confirmed
60-day	76	23000000.0	Low	1.0	0.0	0.0	confirmed
60-day	103	230000.0	Low	0.999	0.001	0.0	confirmed

Note. For each isolate whose day-5 reading sits at the floor, the likelihood of the floor reading under the deposit's own MPN design (duplicate wells, literal lowest-rung pattern) and a log-flat prior give a posterior over the true count, mapped through the isolate's starting density to a posterior over class. An isolate is confirmed when the recorded class reaches 95 per cent, contradicted when another class does, and refused otherwise. The twelve Low labels are confirmed; the six Medium labels are refused, their posterior split across the threshold the

deposit drew at exactly $L/No = 1e-3$. The verdicts are unchanged across both well designs, both readings of the floor value and a uniform prior; under the most conservative analysis (triplicate wells, censoring reading, log-flat prior) the six are contradicted rather than refused (results/tables/exp46_class_posterior.csv).

Table S12. What the assay floor L is in each deposit analysed.

Deposit	States an LOD	States an LOQ	Value used	Units	How obtained	What it should be called
Vijay 2024	no	no	23	MPN per mL	INFERRED	operational assay floor
ERA4TB 2023	no	term only	10, 100 and 400, per plated volume	CFU per mL	DERIVED	minimum reportable positive count
Windels 2024	no	no	none; no value is recoverable	surviving fraction, dimensionless	FLAGGED	no floor is established; a below-limit reading is written as exact zero and fixes no value
Kaur 2024	no	no	none; no floor is used	log10 CFU per mL	NONE	no floor is evidenced; the smallest reading is the smallest reading
Dubey 2026	no	no	10	CFU per mL	DERIVED	minimum reportable positive count

Note. Verdict follows the two kinds of limit defined in Methods: DERIVED means the value follows arithmetically from a recorded plated volume; INFERRED means it was read off the deposit's own behaviour and is labelled as inferred wherever it is used; NONE means no floor is evidenced and none is assumed; FLAGGED means the deposit marks below-limit readings without naming a value, which fixes no floor either.

Table S13. Sensitivity of the load-bearing counts to the choice of floor.

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 3 MPN/mL	isolates short of 4 logs	28
Vijay 2024	floor = 3 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 3 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 3 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 3 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 3 MPN/mL	short of 4 logs, baseline only	17
Vijay 2024	floor = 3.6 MPN/mL	isolates short of 4 logs	28
Vijay 2024	floor = 3.6 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 3.6 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 3.6 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 3.6 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 3.6 MPN/mL	short of 4 logs, baseline only	17
Vijay 2024	floor = 7.2 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 7.2 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 7.2 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 7.2 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 7.2 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 7.2 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 7.4 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 7.4 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 7.4 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 7.4 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 7.4 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 7.4 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 9.2 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 9.2 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 9.2 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 9.2 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 9.2 MPN/mL	calls with several compatible classes	0

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 9.2 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 11 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 11 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 11 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 11 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 11 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 11 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 14 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 14 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 14 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 14 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 14 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 14 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 15 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 15 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 15 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 15 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 15 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 15 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 20 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 20 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 20 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 20 MPN/mL	calls with one compatible class	0
Vijay 2024	floor = 20 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 20 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 21 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 21 MPN/mL	isolates at the floor	0
Vijay 2024	floor = 21 MPN/mL	calls resting on a measured fraction	203
Vijay 2024	floor = 21 MPN/mL	calls with one compatible class	0

Deposit	Floor assumed	Quantity	Value
Vijay 2024	floor = 21 MPN/mL	calls with several compatible classes	0
Vijay 2024	floor = 21 MPN/mL	short of 4 logs, baseline only	21
Vijay 2024	floor = 23 MPN/mL	isolates short of 4 logs	33
Vijay 2024	floor = 23 MPN/mL	isolates at the floor	18
Vijay 2024	floor = 23 MPN/mL	calls resting on a measured fraction	185
Vijay 2024	floor = 23 MPN/mL	calls with one compatible class	12
Vijay 2024	floor = 23 MPN/mL	calls with several compatible classes	6
Vijay 2024	floor = 23 MPN/mL	short of 4 logs, baseline only	21
ERA4TB	per-volume (as analysed)	genuine counts a pooled floor discards (%)	0
ERA4TB	per-volume (as analysed)	below-limit flags contradicted (%)	16.67
ERA4TB	pooled at the most sensitive volume, 100 uL	genuine counts a pooled floor discards (%)	0
ERA4TB	pooled at the most sensitive volume, 100 uL	below-limit flags contradicted (%)	43.17
ERA4TB	pooled at the modal volume, 10 uL	genuine counts a pooled floor discards (%)	1.13
ERA4TB	pooled at the modal volume, 10 uL	below-limit flags contradicted (%)	28.71
ERA4TB	pooled at the least sensitive volume, 2.5 uL	genuine counts a pooled floor discards (%)	5.94
ERA4TB	pooled at the least sensitive volume, 2.5 uL	below-limit flags contradicted (%)	15.26
ERA4TB	pooled at the geometric mean of the per-volume floors	genuine counts a pooled floor discards (%)	1.04
ERA4TB	pooled at the geometric mean of the per-volume floors	below-limit flags contradicted (%)	38.96
Dubey 2026	the below-limit placeholder taken literally	median headroom (log10)	6.08
Dubey 2026	the below-limit placeholder taken literally	cultures short of 4 logs	0
Dubey 2026	100 uL plated, as the deposit's staging record gives	median headroom (log10)	5.08
Dubey 2026	100 uL plated, as the deposit's staging record gives	cultures short of 4 logs	0
Dubey 2026	50 uL plated	median headroom (log10)	4.78
Dubey 2026	50 uL plated	cultures short of 4 logs	0
Dubey 2026	10 uL plated	median headroom (log10)	4.08
Dubey 2026	10 uL plated	cultures short of 4 logs	5
Kaur 2024	100 uL plated, if it had been stated	headroom (log10)	6.03
Kaur 2024	100 uL plated, if it had been stated	4-log endpoint reachable	True
Kaur 2024	the observed minimum, which occurs once	headroom (log10)	5.43

Deposit	Floor assumed	Quantity	Value
Kaur 2024	the observed minimum, which occurs once	4-log endpoint reachable	True
Kaur 2024	10 uL plated, if it had been stated	headroom (log10)	5.03
Kaur 2024	10 uL plated, if it had been stated	4-log endpoint reachable	True
Kaur 2024	2.5 uL plated, if it had been stated	headroom (log10)	4.43
Kaur 2024	2.5 uL plated, if it had been stated	4-log endpoint reachable	True
Windels 2024	no floor is recoverable	readings written as exact zero	6

Note. Each deposit is sensitive to a different thing, so the table is long rather than wide: one row per deposit, floor scenario and quantity. The clinical sweep steps through every three-tube most-probable-number rung at or below 23 per mL, and the number of isolates short of four logs of headroom moves only between 28 and 33 across the whole range, which is why the inferred floor is safe to use. The six-laboratory rows price what pooling the four plating volumes to one floor would cost. The Kaur deposit records no plated volume, so its rows show what assuming one would do: the four-log endpoint stays reachable throughout while the headroom itself moves by 1.6 log10.

Table S14. An MPN floor is not a plate-count floor: the identifiability verdicts of the Results recomputed under the assay's own likelihood.

Panel	Sample day	What the floor reading is taken to be	Floored isolates	One compatible class	Several compatible classes
15-day	5	plate sweep	18	12	
15-day	5	mpn pattern 95	18	12	
15-day	5	mpn pattern 99	18	12	
15-day	5	mpn censor 95	18	12	
15-day	5	mpn censor 99	18	12	
15-day	2	plate sweep	18	2	
15-day	2	mpn pattern 95	18	1	
15-day	2	mpn pattern 99	18	1	
15-day	2	mpn censor 95	18	1	
15-day	2	mpn censor 99	18	1	
60-day	5	plate sweep	6	6	
60-day	5	mpn pattern 95	6	6	
60-day	5	mpn pattern 99	6	6	
60-day	5	mpn censor 95	6	6	
60-day	5	mpn censor 99	6	6	
60-day	2	plate sweep	3	1	
60-day	2	mpn pattern 95	3	1	
60-day	2	mpn pattern 99	3	1	

Panel	Sample day	What the floor reading is taken to be	Floored isolates	One compatible class	Several compatible clas
60-day	2	mpn censor 95	3		1
60-day	2	mpn censor 99	3		1

Note. `plate sweep` is the `[0, L]` interval used in the main text; `mpn pattern` treats the floor reading as the literal lowest-rung well pattern and quotes its two-sided profile interval; `mpn censor` treats it as every pattern whose estimate falls below the lowest rung and quotes the one-sided likelihood bound. The deposit's day-5 splits (12 single, 6 multiple at 15 days; 6 and 0 at 60) are unchanged under every reading at 95 and 99 per cent, because the widest 99 per cent bound, 209 per mL, still falls short of the 230 per mL that would move any of the twelve. At day 2, where the floor is 230, one isolate of eighteen moves from one compatible class to two. The triplicate-well design gives identical verdicts throughout (`results/tables/exp44_mpn_interval_summary.csv`).

Table S15. What is actually known about the assay floor in each deposit.

Deposit	Verdict	Floor used	95% support	Span (log10)	Refuse observability labels
Vijay 2024, clinical isolates (MPN per mL)	INFERRED	23	[9.2, 23.0]	0.398	no
ERA4TB, 100 uL plated (CFU per mL)	DERIVED	10	point mass	0.000	no
ERA4TB, 10 uL plated (CFU per mL)	DERIVED	100	point mass	0.000	no
ERA4TB, 2.5 uL plated (CFU per mL)	DERIVED	400	point mass	0.000	no
Kaur 2024, apramycin grid (log10 CFU per mL)	NONE	-	-	-	yes
Windels 2024, evolved clones (surviving fraction)	NONE	-	-	-	yes
Dubey 2026, hollow fibre (CFU per mL)	DERIVED	10	point mass	0.000	no

Note. And the rule that follows from it. A floor derived from a recorded plated volume is a point mass: one colony in that volume, no inference required. A floor inferred from a pile-up on a most-probable-number rung carries a posterior over the rungs at or below the observed minimum, and the 95 per cent support is quoted. Where no floor is evidenced at all, or where the support spans more than one log10, the observability labels of Section 3 are refused rather than reported -- a label is only as good as the floor it is computed against.

Table S16. The ordinal reanalysis against the linear model it replaces, member for member.

Predictor	Prior culture	Reading day	Linear beta	p (linear)	Survives BH	Odds ratio	p (ordinal)	Survives BH	Sa
time to OD 0.4	15 d	day 5	+0.0266	0.0025	yes	1.096	0.0030	yes	
resistance	15 d	day 5	+0.2589	0.0035	yes	2.317	0.0042	yes	
time to OD 0.4	60 d	day 2	-0.0195	0.0590	no	0.933	0.0470	no	
time to OD 0.4	15 d	day 2	+0.0116	0.0812	no	1.068	0.0618	no	
resistance	15 d	day 2	+0.0515	0.4446	no	1.295	0.4529	no	
time to OD 0.4	60 d	day 5	-0.0041	0.6999	no	0.976	0.4605	no	
resistance	60 d	day 2	-0.0551	0.5486	no	0.836	0.5400	no	
resistance	60 d	day 5	+0.0389	0.6797	no	1.111	0.7150	no	

Note. The linear model scores the ordering 0, 1, 2, which assumes the two class steps are equal; the ordinal model does not. Every direction agrees and the same two members survive correction under both, so the linear treatment did not manufacture the result -- but the coefficients it reports are in a unit that does not exist, which is why the ordinal fit is the one in the main table.

Table S17. The comparison the Methods promise: rows dropped from the association family against rows retained.

Panel	Exclusion	Dropped	Retained	Median log10 N0 retained	Median log10 N0 dropped	p	Resistant, retained
15-day	label missing or 'MDR'	14	202	5.79	5.36	0.198	41%
15-day	growth proxy missing	1	202	5.79	6.79	0.149	41%
60-day	label missing or 'MDR'	20	196	7.36	7.79	0.027	40%
60-day	growth proxy missing	1	196	7.36	7.36	0.749	40%

Note. On starting density and susceptibility. The MDR exclusion differs in susceptibility by construction, since those isolates are outside the resistant-versus-susceptible contrast the family tests. The one difference not by

construction is in the 60-day panel, where the dropped rows sit higher in starting density ($p = 0.027$); it affects 20 rows and no conclusion drawn from that panel.

Table S18. The deepest reduction each assay could resolve.

Dataset	Level of variation	n	Median log10 N0	Assay floor L	delta h (log10)	Fold
ERA4TB, between laboratories at 100 uL	between laboratories, one written protocol	12	4.63	10 CFU/mL, derived from 100 uL plated	2.33	216x
ERA4TB, every laboratory and plating volume	laboratories and plated volumes together	56	4.29	10, 100 or 400 CFU/mL by volume	4.40	25,123x
Vijay, 15-day culture	between clinical isolates	217	5.79	23 MPN/mL, inferred	4.42	26,522x
Vijay, 60-day culture	between clinical isolates	210	7.36	23 MPN/mL, inferred	5.43	269,565x
Kaur, planktonic	technical replicates of one preparation	3	7.03	not stated in the deposit	NA	NA
Kaur, intracellular	technical replicates of one preparation	3	5.94	not stated in the deposit	NA	NA
Dubey, hollow fibre	between cultures, one laboratory	20	6.08	10 CFU/mL, derived from 100 uL plated	0.60	4x

Note. Computed as $h = \log_{10}(N_0/L)$, with the assay floor taken per sample where it varies. Rows compare only within a level of variation: the clinical rows describe between-isolate starting burden, which is biological, and are not a measure of laboratory imprecision. Kaur is NA because its three day-zero readings are technical replicates of one preparation and cannot estimate between-preparation reproducibility, and because that deposit states no quantification limit.

Table S19. Fold below the nominal 0.5 McFarland reference, 1.5e8 CFU/mL.

Dataset	Median log10 N0	Fold below nominal 0.5 McFarland
ERA4TB, between laboratories at 100 uL	4.63	3,525x
ERA4TB, every laboratory and plating volume	4.29	7,751x
Vijay, 15-day culture	5.79	246x
Vijay, 60-day culture	7.36	7x
Kaur, planktonic	7.03	14x
Kaur, intracellular	5.94	170x
Dubey, hollow fibre	6.08	123x

Note. Descriptive only, and not a protocol-compliance metric. A time-kill inoculum is prepared by diluting from a suspension matched to that turbidity, so every entry is expected to sit far below it; the conversion of a turbidity to CFU/mL depends on species, cell aggregation and preparation and is least reliable for mycobacteria. The clinical rows are most probable numbers divided by a reference stated in CFU/mL, so those two ratios cross units and are the least meaningful in the table.

Table S20. Pairs of flasks in the same arm from different laboratories.

Arm	Comparable pairs	Inversions	Rate	95% CI	Called by D/b criterion	Variance: distance	Variance: rate	Dis
All arms pooled	191	66	34.6%	28.1-41.5%	84.3%			
INH 10X MIC						32.3%	67.7%	
INH 1X MIC						17.7%	82.3%	
MXF 10X MIC						35.2%	64.8%	

Note. An inversion is a pair in which the population that fell faster crossed below the assay floor later. 45 further pairs that the censoring could not settle are excluded rather than imputed. The variance columns decompose the spread in crossing time. The rate term is the larger at the flask level in all three arms that can be decomposed, and that ordering is NOT a claim this table supports: the flasks are three to a laboratory and share a starting culture, so the quantity is a between-laboratory statistic on six clusters. Deleting one laboratory sends the distance share above a half in two of the three arms, and a bootstrap over whole laboratories straddles a half in all three. The last two columns are those two checks, and they are the reason the text claims only that the two

terms are of comparable size. Moxifloxacin at one times MIC is not among them: the decomposition is taken on $\log D$ and $\log b$, five of six laboratories record net growth in that arm, and only one flask in it returns a positive fitted rate, so there is no spread in the rate to divide.

Table S21. The same 32-fold concentration range summarised at each sampling day and the concentration slope fitted per interval.

Read at	Survivor ratio (low/high dose)	log10 separation	Slope per doubling	95% interval	p
day 3	6.9x	0.84			
day 7	48.2x	1.68			
day 14	5.2x	0.71			
days 0-3			+0.0590	+0.0327 to +0.0834	0.013
days 3-7			+0.0330	+0.0079 to +0.0586	0.236
days 7-14			-0.0251	-0.0353 to -0.0153	0.091

Note. Upper rows are the day summaries; lower rows the interval slopes. Slopes and intervals are from the replicate-level bootstrap described in the Methods.

Table S22. The six strongest of the 24 comparisons the 217-isolate file supports.

Stratum	Endpoint	n	Spearman rho	p	BH critical value	Survives correction	Resolvable rho
baseline only	MDK99.99 (15d)	162	-0.206	0.0086	0.0021	no	0.15
INH-susceptible	MDK99.99 (15d)	119	-0.218	0.0171	0.0042	no	0.18
INH-susceptible	MDK99 (60d)	117	-0.198	0.0324	0.0063	no	0.18
all isolates	MDK99.99 (60d)	187	-0.153	0.0361	0.0083	no	0.14
INH-susceptible	MDK99.99 (60d)	117	-0.179	0.0540	0.0104	no	0.18
baseline only	MDK99.99 (60d)	156	-0.149	0.0628	0.0125	no	0.16

Note. 4 reach nominal significance where 1.2 are expected by chance; none exceeds its Benjamini-Hochberg critical value, which is ranked against the full family of 24 rather than against any one stratum. MDK99 and MDK99.99 are the minimum durations for a 99 and a 99.99 per cent reduction, the endpoints the text names in words. The final column is the correlation each design could have resolved at 95% confidence.

Table S23. The same nominal concentration expressed in multiples of the MIC each population actually evolved to.

Nominal concentration (ug/mL)	Nutrient levels	Lowest exposure (x MIC)	Highest exposure (x MIC)	Spread	Straddles t
12.5	3	1.49	4.74	3.2x	
25	3	0.55	9.47	17.1x	
50	2	5.08	10.95	2.2x	
100	1	9.72	9.72	1.0x	



Note. At 25 ug/mL the same number denotes a sub-inhibitory exposure in one nutrient condition and a strongly inhibitory one in another.

Table S24. Decomposition of the isoniazid-resistance association with the tolerance class into a mediated and a direct path.

Outcome and stratum	n	Total effect c (95% CI)	Direct effect c' (95% CI)	Mediated effect (95% CI)	Proportion mediat
Linear 0/1/2, all IS/IR	203	+0.256 (+0.087, +0.425)	+0.091 (-0.116, +0.287)	+0.166 (+0.067, +0.279)	6
Linear 0/1/2, baseline isolates	168	+0.226 (+0.038, +0.409)	+0.067 (-0.157, +0.294)	+0.159 (+0.053, +0.295)	7
High versus rest	203	+0.121	+0.013	+0.108 (+0.037, +0.189)	8
Not-low versus low	203	+0.135	+0.077	+0.058 (+0.003, +0.121)	4



Note. Through log10 starting density (mediated) and a direct path, by the product of coefficients with bootstrap percentile intervals. The mediated path excludes zero in every specification and the direct path covers zero in every one. This replaces the percentage attenuation the earlier analysis quoted, which is a descriptive ratio rather than an estimand. Sequential ignorability is assumed and is not testable here; the sensitivity analysis in the Methods reports the residual correlation that would nullify the estimate.

Table S25. What the deposit can and cannot rule out for the isoniazid-resistance seeding gap.

Covariate	Recorded, resistant	Recorded, susceptible	Can it adjust?	Resistance coefficient	Attenuation
INH_MGIT_DST	82/84	0/119	PROXY	-0.851	-0%
RIF_MGIT_DST	82/84	0/119	PROXY	-0.851	-0%
INH_mutation	79/84	1/119	PARTIAL	-0.662	22%
INH_Mykrobe	76/84	0/119	PROXY	-	-
RIF_Mykrobe	76/84	0/119	PROXY	-	-
MIC_INH	67/84	119/119	PARTIAL	-0.963	-13%
MIC_RIF	67/84	119/119	PARTIAL	-0.853	-0%
Time_to_0.4	83/84	119/119	INDEPENDENT	-0.807	5%
Time_point	84/84	119/119	INDEPENDENT	-0.818	4%

Note. Isoniazid-resistant isolates enter this assay ten-fold lower than susceptible ones, and we do not know why. The verdict column is decided on the two recorded-rate columns beside it: PROXY means the covariate is recorded for no isolate in one of the two exposure groups, so its missingness is the exposure; PARTIAL means both groups record it but the missingness is still tied to the exposure (Fisher exact $p < 10^{-6}$); INDEPENDENT means it is not. Seven of the nine covariates are therefore unusable, and they fail in two ways. Five are recorded almost exclusively for resistant isolates: the two susceptibility calls, the two Mykrobe calls and the mutation identity. The susceptibility calls are missing on identical rows, which is why they return identical coefficients, and the Mykrobe calls carry a single value wherever they are recorded in this stratum, so no model can be fitted for them and no coefficient is printed. The other two are the drug MICs, recorded for every susceptible isolate and 67 of 84 resistant ones; the isoniazid MIC separates the two groups completely, so adjusting for it conditions on a graded reading of the exposure. A negative attenuation is an amplification: the adjusted coefficient sits further from zero than the unadjusted one, which is what the isoniazid MIC does at 13 per cent. A coefficient is printed wherever one exists so the circularity is visible, but none of these seven bounds anything. Only 2 covariates are recorded at rates unrelated to susceptibility, and adjusting for either leaves the coefficient within five per cent of its unadjusted value. The file records no referring site and no processing batch, so those cannot be tested at all.

Table S26. What follows a first observed crossing below the assay floor.

Series	n	Visit pairs	P(above to below)	P(below to above)	Never below	One crossing, holds	One crossing, re
all series	360	2232	0.080	0.222	220	56	
INH 10X MIC	72	441	0.137	0.065	34	27	
INH 1X MIC	72	440	0.150	0.373	30	2	
MXF 10X MIC	72	433	0.141	0.175	30	26	
MXF 1X MIC	72	441	0.023	0.923	59	1	
untreated	72	477	0.011	1.000	67	0	

Note. The state below the floor is not absorbing: a series sitting below it reads above again at the next visit with probability 0.22 overall, and that probability rises as drug pressure falls, which is what a plating artefact does. Only 56 of 360 series, 15.6 per cent, show the shape a survival model assumes -- one crossing that holds. The 360 series are four platings of each of 90 flasks and are not independent; the flask-level counts are 53 flasks with a crossing series and 42 with a returning one.

References cited in this supplemental material

- Vijay S, Vinh DN, Hai HT, Ha VTN, Dung VTM, Dinh TD, Nhung HN, Tram TTB, Aldridge BB, Hanh NT, Thu DDA, Phu NH, Thwaites GE, Thuong NTT. 2021. Ribosomal protein S1 is required for growth and antibiotic tolerance in *Mycobacterium tuberculosis*. *Antimicrob Agents Chemother* 65:e00429-21.
- Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study. *Elife* 13:RP93243. <https://doi.org/10.7554/eLife.93243>.
- van Wijk RC, Lucía A, Sudhakar PK, Sonnenkalb L, Gaudin C, Hoffmann E, Dremierre B, Aguilar-Ayala DA, Dal Molin M, Rybniker J, de Giorgi S, Cioetto-Mazzabò L, Segafreddo G, Manganelli R, Degiacomi G, Recchia D, Pasca MR, Simonsson USH, Ramón-García S. 2023. Implementing best practices on data generation and reporting of *Mycobacterium tuberculosis* in vitro assays within the ERA4TB consortium. *iScience* 26:106411. <https://doi.org/10.1016/j.isci.2023.106411>.
- Rabodoarivelo MS, Hoffmann E, Gaudin C, Aguilar-Ayala DA, Galizia J, Sonnenkalb L, Dal Molin M, Cioetto-Mazzabò L, Degiacomi G, Recchia D, Rybniker J, Manganelli R, Pasca MR, Ramón-García S,

- Lucía A. 2025. Protocol to quantify bacterial burden in time-kill assays using colony-forming units and most probable number readouts for *Mycobacterium tuberculosis*. STAR Protoc 6:103643. <https://doi.org/10.1016/j.xpro.2025.103643>.
5. Yamada WM, Schumitzky A, Kryshchenko A, Otalvaro J, Kim S, Louie A, Drusano GL, Neely MN. 2026. Analyzing pharmacodynamic count data that rapidly decrease to zero. CPT Pharmacometrics Syst Pharmacol 15(1). <https://doi.org/10.1002/psp4.70140>.
 6. March VFA, Mchedlishvili K, Goig GA, Maghradze N, Avaliani T, Aspindzelashvili R, Avaliani Z, Kipiani M, Tukvadze N, Jugheli L, Bouaouina S, Doetsch A, Kalkan S, Reinhardt M, Gagneux S, Borrell S. 2025. Within-host evolution of drug tolerance in *Mycobacterium tuberculosis*. bioRxiv 2025.07.29.667394; preprint, not peer reviewed. <https://doi.org/10.1101/2025.07.29.667394>.
 7. Brauner A, Shores N, Fridman O, Balaban NQ. 2017. An experimental framework for quantifying bacterial tolerance. Biophys J 112:2664-2671. <https://doi.org/10.1016/j.bpj.2017.05.014>.
 8. National Committee for Clinical Laboratory Standards. 1999. Methods for determining bactericidal activity of antimicrobial agents; approved guideline. NCCLS document M26-A. National Committee for Clinical Laboratory Standards, Wayne, PA. ISBN 1-56238-384-1.
 9. Balaban NQ, Helaine S, Lewis K, Ackermann M, Aldridge B, Andersson DI, Brynildsen MP, Bumann D, Camilli A, Collins JJ, Dehio C, Fortune S, Ghigo J-M, Hardt W-D, Harms A, Heinemann M, Hung DT, Jenal U, Levin BR, Michiels J, Storz G, Tan M-W, Tenson T, Van Melder L, Zinkernagel A. 2019. Definitions and guidelines for research on antibiotic persistence. Nat Rev Microbiol 17:441-448. <https://doi.org/10.1038/s41579-019-0196-3>.
 10. Vijay S, Bao NLH, Vinh DN, Nhat LTH, Thu DDA, Quang NL, Trieu LPT, Nhung HN, Ha VTN, Thai PVK, Ha DTM, Lan NH, Caws M, Thwaites GE, Javid B, Thuong NT. 2024. Supplementary file 2 to "Rifampicin tolerance and growth fitness among isoniazid-resistant clinical *Mycobacterium tuberculosis* isolates from a longitudinal study" (elife-93243-sup2-v1.xlsx). eLife. <https://doi.org/10.7554/eLife.93243>.
 11. van Wijk RC, Lucía Quintana A, Ramón-García S. 2023. *Mycobacterium tuberculosis* time kill assay data of the standardized protocol within the ERA4TB consortium. figshare. <https://doi.org/10.6084/m9.figshare.19766083.v1>.
 12. Kaur P. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis* — raw data. figshare. <https://doi.org/10.6084/m9.figshare.26462791.v1>.
 13. Kaur P, Ramya VK, Naveenkumar CN, Bharathkumar K, Singh M, Hobbie SN, Shandil RK, Narayanan S. 2024. Apramycin kills replicating and non-replicating *Mycobacterium tuberculosis*. Front Trop Dis 5:1413211. <https://doi.org/10.3389/fitd.2024.1413211>.

14. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence. *ISME J* 18:wrae070. <https://doi.org/10.1093/ismejo/wrae070>.
15. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Source Data to "Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*" (41467_2026_74323_MOESM4_ESM.xlsx). *Nature Communications*. <https://doi.org/10.1038/s41467-026-74323-2>.
16. Dubey V, Darlow C, Gerada A, Unsworth J, Sheth E, Reza N, Farrington N, Haldenby S, Warren D, Liu X, Howard A, Hope W. 2026. Molecular pharmacodynamics of amoxicillin-clavulanic acid for urinary tract infections caused by *Escherichia coli*. *Nat Commun* 17:7504. <https://doi.org/10.1038/s41467-026-74323-2>.
17. van Wijk RC, Solans BP, Chaba L, Sordello S, Upton AM, Nuermberger EL, Robertson GT, Walter ND, Savic RM. 2026. Predicting tuberculosis relapse based on 28-day CFU, RS ratio, and/or drug contribution for novel regimens in the relapsing mouse model. *bioRxiv* 2026.07.27.740024, version 1, posted 30 July 2026; preprint, not peer reviewed. <https://doi.org/10.64898/2026.07.27.740024>.
18. Pym AS, Saint-Joanis B, Cole ST. 2002. Effect of *katG* mutations on the virulence of *Mycobacterium tuberculosis* and the implication for transmission in humans. *Infect Immun* 70:4955-4960. <https://doi.org/10.1128/IAI.70.9.4955-4960.2002>.
19. van Soolingen D, de Haas PEW, van Doorn HR, Kuijper E, Rinder H, Borgdorff MW. 2000. Mutations at amino acid position 315 of the *katG* gene are associated with high-level resistance to isoniazid, other drug resistance, and successful transmission of *Mycobacterium tuberculosis* in the Netherlands. *J Infect Dis* 182:1788-1790. <https://doi.org/10.1086/317598>.
20. Lai RPJ, Ammerman NC, Tasneen R, Almeida DV, Converse PJ, Nuermberger EL. 2023. Using dynamic oral dosing of rifapentine and rifabutin to simulate exposure profiles of long-acting formulations in a mouse model of tuberculosis preventive therapy. *Antimicrob Agents Chemother* 67:e00481-23. <https://doi.org/10.1128/aac.00481-23>.
21. Tabor ST, Friesen AD, Reichlen MJ, Dide-Agossou C, McGrath M, Peterson R, Ganusov VV, Robertson GT, Voskuil MI, Walter ND. 2025. Mind the gap: understanding discordance between culture- and a non-culture-based measure of bacterial burden in murine tuberculosis treatment models. *bioRxiv* posted 18 December 2025; preprint, not peer reviewed. <https://github.com/SamuelTaborCU/Mtb-16S-rRNA-vs-CFU>.
22. McFarland J. 1907. The nephelometer: an instrument for estimating the number of bacteria in suspensions used for calculating the opsonic index and for vaccines. *JAMA* 49:1176. <https://doi.org/10.1001/jama.1907.25320140022001f>.

23. Franzblau SG, DeGroot MA, Cho SH, Andries K, Nuermberger E, Orme IM, Mdluli K, Angulo-Barturen I, Dick T, Dartois V, Lenaerts AJ. 2012. Comprehensive analysis of methods used for the evaluation of compounds against *Mycobacterium tuberculosis*. *Tuberculosis (Edinb)* 92:453-488.
<https://doi.org/10.1016/j.tube.2012.07.003>.
24. Brauner A, Fridman O, Gefen O, Balaban NQ. 2016. Distinguishing between resistance, tolerance and persistence to antibiotic treatment. *Nat Rev Microbiol* 14:320-330.
<https://doi.org/10.1038/nrmicro.2016.34>.
25. Windels EM, Cool L, Persy E, Swinnen J, Matthay P, Van den Bergh B, Wenseleers T, Michiels J. 2024. Data supporting "Antibiotic dose and nutrient availability differentially drive the evolution of antibiotic resistance and persistence". Zenodo. <https://doi.org/10.5281/zenodo.7550302>.
26. Fisher RA. 1935. The logic of inductive inference. *J R Stat Soc* 98:39-82.
<https://doi.org/10.2307/2342435>.
27. McNemar Q. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12:153-157. <https://doi.org/10.1007/BF02295996>.
28. Wilcoxon F. 1945. Individual comparisons by ranking methods. *Biom Bull* 1:80.
<https://doi.org/10.2307/3001968>.
29. Turnbull BW. 1976. The empirical distribution function with arbitrarily grouped, censored and truncated data. *J R Stat Soc Series B Stat Methodol* 38:290-295. <https://doi.org/10.1111/j.2517-6161.1976.tb01597.x>.